A SIGNED-DIRECTION INVARIANT FOR TRIANGLE TILINGS, AND THE EXCLUSION OF PRIMES CONGRUENT TO 3 MODULO 4

VICO BONFIOLI

Abstract. For a triangle T and a triangle ABC, an N*-tiling* is a dissection of ABC into N triangles each congruent to T. Erdős asked which N occur; a folklore conjecture is that no prime $N \equiv 3 \pmod 4$ with $N > 3$ does (the value $N = 3$ occurs). By the classification of Laczkovich and the branch theorems of Beeson, the conjecture reduces to a tile with a $2\pi/3$ angle on a non-equilateral triangle, where a prime count forces ABC isosceles. For that isosceles case Beeson ruled out $N < 36$ and explicitly left open whether N can be prime (the smallest *known* tiling has 2673 tiles); we settle it. We introduce a translation-invariant, signed-direction tiling functional, linear in edge length and hence insensitive to non-edge-to-edge incidences, whose integrality forces, for an isosceles target, that $(c-a-b)/\sqrt{b}$ be an integer for a primitive triple with $c^2 = a^2 + ab + b^2$; an elementary lemma shows this never occurs. With a self-contained reduction of the scalene shapes, this excludes every prime $\equiv 3 \pmod 4$ exceeding 3 *except* the base-β candidates $N = 3f^2 - e^2$, conditional on the cited classification of the remaining branches. Those candidates are exactly the primes $\equiv 11 \pmod{12}$ (Theorem 2), so the exclusion is unconditional on the class $7 \pmod{12}$ — half of all primes $\equiv 3 \pmod 4$ by Dirichlet, and the class containing 19. The class $11 \pmod{12}$ remains open in general; its members are settled individually by exact search, and 42 of them lie below 1000, of which six are so far resolved. Beyond primes, we determine the admissible spectrum of each sporadic $2\pi/3$ branch — for the isosceles target, writing $b = de^2$ with d squarefree, every count has the form $N = dw^2(a + 2b)$ with $e \mid w(c - a - b)$ — bounding the realizable set from outside; and a spectrum theorem shows that the counts passing all invariant conditions on the isosceles and F_1 targets are *exactly* Zhang's constructed families, closing the necessary side of those branches; the equilateral criteria likewise reduce to elementary divisor conditions. Combining the spectra with the published branch theorems and an exhaustive exact search of the surviving finite instances, we settle every previously undetermined value up to 80 with one pending exception: no triangle can be cut into 14, 15, 21, 22, 30, 33, 35, 38, 39, 42, 51, 55, 56, 57, 60, 63, 69, or 76 congruent triangles (46, 62, 78 likewise, and 79 — a prime — by the main theorem); triangles *can* be cut into 44, 77 (by Beeson's four-component construction), and 80 (a sum of two squares) congruent triangles; the values 66 and 70, previously excluded via theorems of [2] whose proofs we show to be unsound, are undetermined pending direct exhaustive searches of their single surviving instances. The 44-tiling is exhibited explicitly — the isosceles $(16, 16, 22)$ cut into 44 copies of the $(2, 3, 4)$ tile, found by search and verified exactly, to our knowledge the first tiling in the isosceles $3\alpha + 2\beta = \pi$ cases and the smallest known tiling in any incommensurable branch. Membership in the tile-count set is shown to be decidable. The arithmetic and combinatorial layer is machine-checked in Lean 4 with Mathlib.

1. Introduction

Let T be a triangle (the *tile*). An N*-tiling* of a triangle ABC is a way of writing ABC as a union of N triangles, each congruent to T, with pairwise disjoint interiors. We do *not* assume the tiling is edge-to-edge: a vertex of one tile may lie in the relative interior of an edge of another. Erdős asked (reported by Soifer [13]) to determine the set of N for which some triangle admits an N-tiling by some tile. Every perfect square occurs (any triangle, by the n^2 subdivision), and Soifer observed that $n^2 + m^2$, $2n^2$, $3n^2$ and $6n^2$ occur as well; in particular $3 = 3 \cdot 1^2$ occurs.

Date: July 19, 2026.

The outstanding question is whether the primes $\equiv 3 \pmod 4$ that exceed 3, which are exactly the odd primes that are neither sums of two squares nor of the form $3n^2$, are excluded.

Theorem 1. *Let $N > 3$ be a prime with $N \equiv 3 \pmod 4$ which is* not *a base-β candidate — that is, $N \neq 3f^2 - e^2$ for every coprime pair $1 \le e < f$. Then N is not a number of congruent triangles into which some triangle can be cut. In particular, since 19 is not of that form, no triangle can be cut into 19 congruent triangles.*

Theorem 2 (The exceptional set is a congruence class)**.** *Let $p > 3$ be prime with $p = 3f^2 - e^2$ for some coprime e, f. Then $p \equiv 11 \pmod{12}$. Conversely every prime $p \equiv 11 \pmod{12}$ is of that form. Hence the base-β candidates among the primes are exactly the primes $\equiv 11 \pmod{12}$.*

Proof. ($\Rightarrow$) If $3 \mid e$ then $3 \mid p$, impossible for a prime $p > 3$; so $e^2 \equiv 1 \pmod 3$ and $p \equiv -1 \equiv 2 \pmod 3$. Next, e, f are not both even (coprimality) and not both odd, since then $p = 3f^2 - e^2 \equiv 3 - 1 = 2 \pmod 4$ would be an even prime exceeding 3; in either remaining case $p \equiv 3 \pmod 4$. Combining, $p \equiv 11 \pmod{12}$. ($\Leftarrow$) For $p \equiv 11 \pmod{12}$ we have $p \equiv 3 \pmod 4$ and $p \equiv 2 \pmod 3$, so by reciprocity $\left(\frac{3}{p}\right) = 1$ and p splits in $\mathbb{Q}(\sqrt{3})$. The discriminant-12 form class group has order two, represented by $x^2 - 3y^2$ and $3y^2 - x^2$; the first forces $p \equiv 1 \pmod 3$ and the second $p \equiv 2 \pmod 3$, so p is represented by $3y^2 - x^2$. Primitivity is automatic for prime p. $\square$

Corollary 3 (Theorem 1, congruence form)**.** *No prime $p \equiv 7 \pmod{12}$ is a number of congruent triangles into which a triangle can be cut. Equivalently: a prime $p > 3$ with $p \equiv 3 \pmod 4$ is excluded unconditionally unless $p \equiv 11 \pmod{12}$.*

Proof. A prime $p > 3$ with $p \equiv 3 \pmod 4$ satisfies $p \equiv 7$ or $p \equiv 11 \pmod{12}$, the class $3 \pmod{12}$ forcing $3 \mid p$. If $p \equiv 7 \pmod{12}$ then by Theorem 2 it is not a base-β candidate, so Theorem 1 applies with no exception. $\square$

Remark 4 (Why the forms, and what the reformulation buys)**.** Every tile count in the classification is a value of a binary quadratic form: $e^2 + f^2$ (discriminant -4) on the commensurable branch, the Eisenstein norm $a^2 + ab + b^2$ (discriminant -3) for a $2\pi/3$ tile, $3f^2 - e^2$ (discriminant 12) and $2f^2 - e^2$ (discriminant 8) on the $3\alpha + 2\beta$ branch. The classical exclusion of primes $\equiv 3 \pmod 4$ on the commensurable branch is exactly Fermat's two-squares theorem for the first form; Theorem 2 is its analogue for the third. Two consequences. First, the exception in Theorem 1 is not a Diophantine side condition but a single arithmetic progression, and by Dirichlet it carries half of the primes $\equiv 3 \pmod 4$: the theorem is unconditional on the other half. Second, $19 \equiv 7 \pmod{12}$, so the value in Erdős's question lies in the unconditional class. The forward implication is machine-checked (`basebeta_prime_mod_twelve`, `BaseBetaMod12.lean`); it is the direction the corollary uses. The converse was checked for all 109 primes $\equiv 11 \pmod{12}$ below 3000 with no exception.

Remark 5 (The quoted form of [2, Lem. 6] is false)**.** That lemma is usually quoted as "every side of ABC carries at least two c-edges". In that form it is false. Take the tile $(2, 3, 4)$ and let $ABC = (4, 6, 8)$, its double; the four-fold midpoint subdivision is a genuine tiling by four congruent copies, and its sides decompose as b^2, c^2, a^2 — two of the three carry *no* c-edge. Here ABC is similar to the tile, a case treated separately in [2], so the true statement must carry hypotheses that the usual quotation omits. We therefore do not use it: Proposition 47 now invokes only the γ-trap (at least *one* c-edge per side), which is proved here from the vertex figures and machine-checked, and which suffices because the base length identity gives $Y \le (g - M)b$ unconditionally — a single c-edge already contradicts. An exhaustive check over all 11,338 admissible pairs (N, M) with $N < 2500$ prime found *no* decomposition of the base containing even one c-edge, so the margin is not delicate; the citation [2, Lem. 45(iii)] is likewise not needed for the conclusion.

Remark 6 (The scope of Theorem 1)**.** The base-β exception is essential. The base-β candidates are exactly the $N = 3f^2 - e^2$ with $\gcd(e, f) = 1$, $1 \le e < f$. Those with e, f of opposite parity are $\equiv 3 \pmod 4$; those with e, f both odd are $\equiv 2 \pmod 4$, hence even, so among the *primes* only the former occur — and by Theorem 2 they are exactly the primes $\equiv 11 \pmod{12}$. (Composite candidates of either parity occur and appear in Table 1: 26, 66 and 74 are $\equiv 2 \pmod 4$.) The smallest prime candidates are

$$11,\ 23,\ 47,\ 59,\ 71,\ 83,\ 107,\ 131,\ 167,\ 179,\ 191,\ \ldots$$

For each, the tile $(ef,\ f^2 - e^2,\ f^2)$ and the target $(f^3, f^3, e(3f^2 - e^2))$ pass every *sound* necessary condition (Remark 9); the only published exclusion is [2, Thm. 14], whose $g \mid M$ is not merely unproven but **false** (Remark 33), so we cannot invoke it, and no general replacement is known (Remark 45). The theorem therefore cannot drop the exception without appealing to that false citation. The candidates are settled one at a time by exhaustive search (Table 1): $11, 23, 47, 59, 71, 107$ are excluded and 83 remains under search. Thus Theorem 1 is unconditional for every prime $N \equiv 3 \pmod 4$, $N > 3$, outside the sequence above, and for $11, 23, 47, 71, 107$ besides. The principal consequence is unaffected: $19 \ne 3f^2 - e^2$, and the two frontier claims of Section 7 rest only on candidates that have been searched.

The restriction $N > 3$ is necessary: an equilateral triangle joined to its centre splits into 3 congruent triangles, so $N = 3$ occurs.

The proof rests on Laczkovich's classification of triangle tilings [9, 10], organized by Beeson across a series of papers [1, 2, 3, 4, 5]. For a prime number of tiles every branch of the classification is excluded by an existing theorem, with one exception: a tile having a $2\pi/3$ angle, tiling a non-equilateral triangle. We treat that branch.

Zhang [7] classifies the non-equilateral $2\pi/3$ shapes into six sporadic similarity types (Section 2 below), constructs families of tilings for each, and conjectures that the constructed N, all composite, are the only ones; the present results prove the no-prime half of that conjecture for the isosceles type, and reduce the scalene types directly. For the isosceles type Beeson [3] proved that no tiling exists with $N < 36$ and *explicitly left open whether N can be prime*; the smallest tiling presently known has 2673 tiles (Herdt, reported in [3]). The invariant below settles the question, excluding *every* prime. The adjacent Erdős Problem 633, on tilings into a square number of triangles, was recently settled by Beeson, Laczkovich and Zhang [8].

Two reductions prepare it. First, such a tile has commensurable sides, so after scaling we may take it integer-sided with $c^2 = a^2 + ab + b^2$; for the *isosceles* targets to which the theorem reduces, this is Beeson's *accepted* isosceles paper ([3, Thm. 12.5]: any isosceles triangle tiled by a $2\pi/3$-tile with α/π irrational forces the tile rational), so that citation covers the isosceles targets; but the reduction to an isosceles target (Section 3) already works with an integer-sided tile, so the rationality input actually used in that order is the general-target theorem of Beeson–Zhang, a preprint — we record this plainly rather than claim a refereed input; the general-tile rationality of Beeson–Zhang [6] gives the same uniformly. Second, an elementary area-and-incidence argument shows that a *prime* number of such tiles forces ABC to be *isosceles* (Section 3).

The main construction is in Section 4. The natural parity argument for a $2\pi/3$ tile (a black/white colouring of the tiles) fails, because a vertex of the tiling may be surrounded by an even number of tiles, so no consistent two-colouring exists. We replace it by a functional Φ that assigns to a directed edge of direction $\theta = j\frac{\pi}{3} + k\alpha$ the weight $(\text{length}) \cdot (-1)^j$. Two features make it work: $\Phi(\theta+\pi) = -\Phi(\theta)$, so interior edges cancel; and the weight is *linear in edge length*, so the cancellation is unaffected by non-edge-to-edge incidences, where one long edge meets several shorter ones summing to it. Summing over the tiles, Φ of the boundary equals an integer multiple of a fixed tile-value $c+a-b$. For an isosceles target this forces $(c-a-b)/\sqrt{b} \in \mathbb{Z}$, which never holds (Section 5). Assembling the branches proves Theorem 1 (Section 6).

The combinatorial claims are verified by the scripts in the accompanying repository, and the entire arithmetic layer, including Lemma 54 and the reduction of Section 3, is formalized and machine-checked in Lean 4 with Mathlib; see Section 8.

2. Preliminaries

The classification. Fix a prime $N = p > 3$ with $p \equiv 3 \pmod 4$. By Laczkovich [9, 10], every N-tiling of a triangle falls into one of finitely many types of pairs (ABC, T). The relevant dichotomy, in Beeson's organization [4], is: either the angles of T are pairwise commensurable with π, or they are not. If they are commensurable, then N is a square, a sum of two squares, or 2, 3 or 6 times a square [4, Thm. 3 and Table 3]; but $p > 3$ with $p \equiv 3 \pmod 4$ is a square only if p is one (it is not), a sum of two squares only if $p \not\equiv 3 \pmod 4$, and $2n^2, 3n^2, 6n^2$ only if p is even or $3 \mid p$ (here p is odd and $p > 3$), so p is none of these (it is precisely for $p = 3 = 3 \cdot 1^2$ that this step would fail). If the angles are not commensurable, then, after the right-angle and similar-tile cases, which give N a sum of two squares or $N = n^2$ [1, 12], the tile has a special angle and falls into one of: $3\alpha + 2\beta = \pi$, for which N is not prime (originally [2]; see Remark 7 and Propositions 47 and 8 — the arithmetic core is machine-checked in Lean for *four* of the five target shapes, with our own proofs replacing Beeson's Theorems 18 and 20, whose printed arguments are unsound; the fifth, the base-β isosceles target, is the exception, treated in Remark 9); an isosceles tile with $\gamma = 2\alpha$, for which N is not squarefree [3, Thm. 11.7]; or the tile has an angle equal to $\pi/3$ or $2\pi/3$. The case of a $\pi/3$ or $2\pi/3$ tile on an *equilateral* triangle gives N not prime [5]. What remains for a prime N is a tile with a $\pi/3$ or $2\pi/3$ angle on a *non-equilateral* triangle. The vertex enumeration of Section 2 shows the $\pi/3$ case admits only equilateral or tile-similar ABC, already covered; so the sole remaining branch is a $2\pi/3$ tile, which we now fix.

Remark 7 (The $3\alpha + 2\beta = \pi$ prime exclusion)**.** The step above invokes [2] for the claim that a $3\alpha + 2\beta = \pi$ tiling has N composite. For the two scalene targets $(2\alpha, \beta, \alpha+\beta)$ and $(2\alpha, \alpha, 2\beta)$ this is Beeson's Theorems 8 and 12, whose arithmetic cores we machine-check in Lean (`lean/Beeson3NotPrime.lean`): the first tiling equation $N + M^2 = 2K^2$ with $K \mid N$ admits no odd prime, and the second forces $N = (2f^2 - e^2)(3f^2 - e^2)k^2$ — a product of two factors exceeding 1, hence composite. The isosceles target with base angles $\alpha+\beta$ is Beeson's Theorem 18, but its published proof is *unsound*: under Beeson's scaling $a = N - M^2,\ c = N + M^2,\ b = c - a^2/c$ one has $bc = c^2 - a^2 = 4NM^2$, so $c^2 b = 4NM^2(N + M^2) \equiv 0 \pmod N$ *identically* (no primality used), not the $M^4(M^2+1)$ his argument needs — a dropped factor of a; the residue is already false on genuine composite tilings ($N = 48,\ M = 4$ gives $c^2 b \equiv 0$ while $M^4(M^2+1) \equiv 32$), so the step forcing $N = M^2 + 1$ is vacuous. The conclusion is nonetheless correct, by the base-length obstruction of Proposition 47.

Two further defects of [2] affect the base-α target. First, the squarefree assertion of its Lemma 8 ("$g = \gcd(a, c)$ is squarefree") is *false*: the tile $(4, 15, 16)$ — a genuine $3\alpha + 2\beta$ tile, $e = 1,\ f = 4$ — has $g = 4$ (its proof breaks at the inference $p^2 \mid a \Rightarrow p^2 \mid \hat{a}$); the parametrized family $(ef,\ f^2 - e^2,\ f^2)$ realizes *every* f coprime to e, squarefree or not. Second, Theorem 19's claim $g \mid M$ rests on that lemma *and* on the assertion that $bc^3(a + c)$ is not divisible by g^5 — but with $c = g^2$ it carries g^7 identically, so the argument collapses, and with it the proof of the base-α prime exclusion (Theorem 20). Proposition 8 repairs the latter without any appeal to $g \mid M$.

Proposition 8 (Base-α prime exclusion, replacing [2, Thm. 20])**.** *No isosceles triangle with base angles α, where $3\alpha + 2\beta = \pi$ and $\alpha \notin \mathbb{Q}\pi$, is N-tiled for N prime.*

Proof. Write $s = e/f$ in lowest terms, so the tile is $(ef,\ f^2 - e^2,\ f^2)$. Two necessary conditions suffice: the tiling equation $N(f-e)(2f+e)^2 = M^2(f+e)(2f^2-e^2)$ (the sound half of [2, Thm. 19]) and the integrality of the equal side X, where $X^2 = Nbc/(2-s^2)$. Substituting the tiling equation into X^2 gives the exact value $X = M(f+e)f^2/(2f+e)$; as $\gcd(2f+e, f) = \gcd(2f+e, f+e) = 1$, integrality forces $(2f+e) \mid M$. Writing $M = (2f+e)m$ reduces the tiling equation to

$$N(f-e) \;=\; m^2\,(f+e)\,(2f^2-e^2).$$

Put $d = f - e$ and $Q = 2f^2 - e^2 = e^2 + 4ed + 2d^2$. Then $\gcd(d, Q) = 1$ (a common prime would divide e^2, hence e, hence $\gcd(e, f)$), so the minimal prime factor of $Q \geq 7$ divides N and equals N; writing $Q = NQ_1$ and cancelling N leaves $d = m^2(f+e)Q_1 \geq f+e > f-e = d$, a contradiction. Both arithmetic pillars are machine-checked in Lean (`lean/IsoAlphaPrime.lean`, axiom-clean: `isoalpha_X_forces`, `isoalpha_not_prime`). □

Remark 9 (The base-β target)**.** The fifth target, isosceles with base angles β, is the one shape where we cannot replace Beeson's argument with a sound one. Beeson's Theorem 14 also asserts $g \mid M$, by a proof of the same broken shape as Theorem 19's (it claims abc^5M^2 is not divisible by g^7, but with $c = g^2$ it carries g^{11}). And here the defect *bites*: the sound necessary conditions do not exclude prime N. With $s = e/f$ in lowest terms the tile is $(ef,\ f^2 - e^2,\ f^2)$, the equal side is $X = f^3M/(f+e)$ and the base is $Y = eM(3f^2 - e^2)/(f+e)$; integrality of X forces $(f+e) \mid M$, and writing $M = (f+e)m$ the Theorem-14 equation $N(e+f)^2 = M^2(3f^2 - e^2)$ collapses to

$$N = m^2\,(3f^2 - e^2),$$

which *is* prime for $m = 1$ and $3f^2 - e^2$ prime (e.g. $N = 11, 23, 47, 59, 71, \ldots$, always $\equiv 3 \pmod 4$). Each such N passes every sound necessary condition ($X, Y \in \mathbb{Z}$, parity, range) and is killed only by the unsound $g \mid M$. Unlike the base-α case (Proposition 8), no arithmetic contradiction follows; *a general proof that these candidates do not tile is open*, and to our knowledge is not supplied by a correct argument in the literature. For the finitely many instances that bear on the values settled in this paper we resort to the direct corner-anchored search. An exhaustive audit of $N \leq 80$ — computing the equal side X and base Y exactly and retaining every candidate that passes all *sound* necessary conditions ($X, Y \in \mathbb{Z}$, parity, range) — shows the affected set is precisely

$$\text{primes } \{11, 23, 47, 59, 71\} \text{ (base-}\beta) \quad \text{and} \quad \text{composites } \{21, 70\} \text{ (base-}\alpha),\ \{39, 66\} \text{ (base-}\beta).$$

The engine settles them one at a time; all searches are exact and exhaustive:

N	branch	tile	target	nodes
11	base-β	$(2,3,4)$	$(8,8,11)$	135
21	base-α	$(2,3,4)$	$(12,12,21)$	13 659
23	base-β	$(6,5,9)$	$(27,27,46)$	19 677
47	base-β	$(4,15,16)$	$(64,64,47)$	12 440
39	base-β	$(12,7,16)$	$(64,64,117)$	825 724
71	base-β	$(10,21,25)$	$(125,125,142)$	553 417

each returning EXHAUSTED, NO TILING. The remaining three — $N = 59$ (base-β, $(20, 9, 25)$ on $(125, 125, 236)$), $N = 66$ (base-β, $(15, 16, 25)$ on $(125, 125, 198)$) and $N = 70$ (base-α, Remark 79) — are still searching, and their values are treated as pending. For the base-β target the $3\alpha + 2\beta$ prime argument rests on either [2, Thm. 14] (whose printed proof is unsound) or a per-value search, *not* on a machine-checked general theorem; this is the one remaining gap in the prime dichotomy. Theorem 10 removes $N = 11$ unconditionally, Remark 11 isolates the $e = 1,\ f \geq 3$ structure, and Remark 32 explains why the method [2] used cannot repair the rest.

Theorem 10 (The base-β family at $e = 1$, $f = 2$)**.** *The isosceles triangle with base angles β and sides $(8, 8, 11)$ admits no tiling by $N = 11$ copies of the tile $(2, 3, 4)$. Consequently the base-β branch is closed for $N = 11$, unconditionally.*

Proof. Write $\omega = \alpha/2$. The law of cosines gives $\cos\alpha = (2f^2 - e^2)/(2f^2)$, hence $\sin\omega = e/(2f)$; as $1 \le e < f$ places this strictly in $(0, \frac{1}{2})$, *Niven's theorem* forces α/π irrational — for every (e, f), with no appeal to [9] (`tile_alpha_irrational`). Irrationality pins the vertex figures: a vertex $x\alpha + y\beta + z\gamma = k\pi$ forces $y + z = 2k$ and $2x + z = 3y$, so a π-vertex is $\{3\alpha, 2\beta\}$ or $\{\alpha, \beta, \gamma\}$, each base corner carries *exactly one* tile (its β-vertex) and the apex *exactly three* (`vertex_pi`, `vertex_beta_corner`, `vertex_apex`); in particular a π-vertex carries at most one γ, since $2\gamma = \pi + \alpha > \pi$. Each side of ABC is partitioned into whole tile edges (a supporting line meets a convex tile in a face).

Two citation-free constraints now cut the base. (a) *Corner parallelogram*: the unique β-corner tile T_B has its b-edge a chord $[P, Q]$, matched (by the mod-f edge arithmetic) by a single tile T' whose α, γ are swapped, so $T_B \cup T'$ is a parallelogram with sides a, c; hence P, Q are $(1, 1, 1)$-vertices whose third tile carries β, and the second base edge from each corner lies in $\{a, c\}$. So a b-edge can occupy only positions $3, \ldots, k - 2$ of a base of k edges. (b) *γ-trap*: no boundary junction carries γ on both sides, and only a c-edge leaves the γ-state, so every side carries at least one c-edge.

At $e = 1$, $m = 1$ the base $Y = 3f^2 - 1$ has $q_{\text{base}} \equiv 1 \pmod{f}$, and the only compositions are $B_0 = \{b, c, c\}$ $(k = 3)$, $B_1 = (a^f, b, c)$ $(k = f + 2)$ and $B_2 = (a^{2f}, b)$ $(k = 2f + 1)$. B_2 dies by (b). B_0 dies by (a) for *every* f: its single b would need a position in $3, \ldots, 1$. At $f = 2$, B_1 also dies by (a): $k = 4$ leaves positions $3, \ldots, 2$. So no base survives, and $N = 11$ admits no tiling. □

Remark 11 (The $e = 1$, $f \ge 3$ cases are *not* settled)**.** For $f \ge 3$ the composition $B_1 = (a^f, b, c)$ survives every citation-free constraint above, and it is then the *unique* possible base. Two consequences follow. First, B_1 carries exactly *one* c-edge; so [2, Lem. 6] ("every side carries at least two c-edges") removes it at a stroke — but that citation is doing *all* of the work, and any tiling of this family would *refute* Lemma 6 rather than be excluded by it, so invoking it is tantamount to assuming the conclusion. Given that [2] is the source whose Lemma 8 we show false, whose Theorems 18–20 we show unsound, and whose Theorem 14 we show to be outright *false* (Remark 33), we do not invoke it. Second, our own forcing argument targets B_0, which cannot occur in any case, so it says nothing about $f \ge 3$. We therefore record $N = 26, 47, 74, 107, 146, 191, \ldots$ as *open* in this branch, with $N = 26, 47, 74$ independently excluded by exhaustive search.

What survives citation-free is a sharp structure theorem: *if* an $e = 1$, $m = 1$, $f \ge 3$ base-β tiling exists, then $q_{\text{equal}} = 0$, its base is exactly (a^f, b, c) with $E_1 = E_k = a$ and the c-edge interior, both equal sides begin and end with c, and every base junction is a $(1, 1, 1)$-vertex.

The next theorem replaces the $e = 1$ case analysis by a classification valid for *every* e, and shows why $m = 1$ — the prime case — is the rigid one. It is machine-checked (`BaseBetaWalks.lean`, axiom-clean).

Theorem 12 (Boundary walks of the base-β target at $m = 1$)**.** *Let $(a, b, c) = (ef,\ f^2 - e^2,\ f^2)$ be the primitive $3\alpha + 2\beta$ tile ($1 \le e < f$, $\gcd(e, f) = 1$) and let $m = 1$, so that the target is $(f^3, f^3, e(3f^2 - e^2))$ and $N = 3f^2 - e^2$. Write a walk along a side as $P \cdot a + Q \cdot b + R \cdot c$ equal to that side's length. Then both walk sets are cut out by the* same *linear form $\langle e, f, 1\rangle$, the base at level $2e$ and the equal side at level f:*

$$\begin{aligned} &\textit{base:} \quad (P, Q, R) = (je + fp,\ e + fj,\ R), && pe + jf + R = 2e, && j \in \mathbb{Z}_{\ge 0}, \\ &\textit{side:} \quad (P, Q, R) = (qe + fp,\ fq,\ R), && pe + qf + R = f, && q \in \mathbb{Z}_{\ge 0}, \end{aligned}$$

with $p \in \mathbb{Z}$ and $P, R \ge 0$. In the thin regime $f > 2e$ the levels force short lists:

(i) (trichotomy) *the base is exactly one of* $\{b^e, c^{2e}\}$, $\{a^f, b^e, c^e\}$, $\{a^{2f}, b^e\}$; *in particular it carries* exactly e *b-edges;*

(ii) (dichotomy) *each equal side is* $\{a^e, b^f\}$ *or* $\{a^{fp}, c^{f-pe}\}$ *for some* $p \geq 0$.

Feeding in the γ*-trap* ($R \geq 1$ *on every side, Remark 45*) *removes the thinness hypothesis altogether. For* every (e, f) *at* $m = 1$*:*

(iii) no equal side carries a b-edge — *each is* $\{a^{fp}, c^{f-pe}\}$, $p \geq 0$; *so* all *boundary b-edges lie on the base;*

(iv) *the base's b-count* $Q = e + fj$ *obeys* $j(f - e) \leq e - 1$.

In particular $e = 1$ *forces* $j = 0$ *for every* f — the base carries exactly one b-edge — *and* $f > 2e$ *forces* $j = 0$, *recovering* (i) *minus its* $R = 0$ *member.*

Proof. Base. Reducing $P \cdot ef + Q(f^2 - e^2) + Rf^2 = e(3f^2 - e^2) \bmod f$ gives $Qe^2 \equiv e^3$, so $f \mid Q - e$ by $\gcd(e, f) = 1$; as $0 \leq Q$ and $0 < e < f$, the least nonnegative residue is e, whence $Q = e + fj$ with $j \geq 0$. Substituting and cancelling one factor f leaves the *halved equation*

$$Pe + j(f^2 - e^2) + Rf = 2ef. \tag{1}$$

Reducing (1) mod f gives $e(P - je) \equiv 0$, so $f \mid P - je$, say $P = je + fp$, $p \in \mathbb{Z}$; substituting into (1) and cancelling f gives $pe + jf + R = 2e$. Conversely every such triple solves the original equation. For $f > 2e$: from (1), $j(f^2 - e^2) \leq 2ef$; if $j \geq 2$ then $f^2 - e^2 \leq ef$, i.e. $f^2 - ef - e^2 \leq 0$, whereas $(f - 2e)(f + e) = f^2 - ef - 2e^2 > 0$ gives $f^2 - ef - e^2 > e^2 > 0$ — contradiction. If $j = 1$ then $f \mid P - e$ forces $P \geq e$, and (1) gives $e^2 + (f^2 - e^2) \leq 2ef$, i.e. $f^2 \leq 2ef$, i.e. $f \leq 2e$ — contradiction. So $j = 0$, whence $f \mid P$, $P = fp$ with $p \geq 0$, and $pe + R = 2e$ forces $p \in \{0, 1, 2\}$ and $R = e(2 - p)$, giving the three walks.

Side. The same reduction on $P \cdot ef + Q(f^2 - e^2) + Rf^2 = f^3$ gives $Qe^2 \equiv 0$, so $f \mid Q$, $Q = fq$; cancelling f leaves $Pe + q(f^2 - e^2) + Rf = f^2$, whose reduction mod f gives $f \mid P - qe$, say $P = qe + fp$, and cancelling f again gives $pe + qf + R = f$. For $f > 2e$: $q \geq 2$ would give $2(f^2 - e^2) \leq f^2$, i.e. $f^2 \leq 2e^2 < f^2$, absurd. If $q = 1$ then $Pe + Rf = e^2$ and $f \mid P - e$ force $P = e$, $R = 0$; if $q = 0$ then $f \mid P$, $P = fp$ and $pe + R = f$.

(iii) and (iv) are one computation, and need no thinness. Suppose a side walk has $q \geq 1$. From $P = qe + fp \geq 0$ we get $fp \geq -qe > -qf$ (using $e < f$ and $q \geq 1$), so $p > -q$, i.e. $p \geq 1 - q$. The level equation then gives

$$R \ = \ f - pe - qf \ \leq \ f - (1 - q)e - qf \ = \ (1 - q)(f - e) \ \leq \ 0,$$

contradicting $R \geq 1$. Hence $q = 0$, which is (iii). Identically for the base with $P = je + fp \geq 0$: if $j \geq 1$ then $p \geq 1 - j$ and

$$R \ = \ 2e - pe - jf \ \leq \ 2e - (1 - j)e - jf \ = \ e - j(f - e),$$

so $R \geq 1$ forces $j(f - e) \leq e - 1$, which is (iv) (and it holds trivially at $j = 0$ since $e \geq 1$). For $e = 1$ this reads $j(f - 1) \leq 0$, so $j = 0$; for $f > 2e$ we get $je < j(f - e) \leq e - 1$, so $j = 0$. ☐

Theorem 13 (Apex mismatch: the pierced corner)**.** *Let a base-*β *target be tiled at any scale* m *such that both equal sides end (at the apex) with a c-edge — automatic at* $m = 1$*, where the sides are b-free. Then, writing* T_2 *for the middle of the three apex tiles:*

(i) *the outer apex tiles carry c on the boundary and b inward, and* T_2 *carries b and c inward, so exactly one inner apex ray carries* T_2*'s c-edge against a neighbour's b-edge: a T-junction at distance b from the apex, whose vertex figure on the neighbour's side is* $(1, 1, 1)$ *(one* α*, one* β*, plus the neighbour's* γ*);*

(ii) *the remaining segment of* T_2*'s c-edge has length* $c - b = e^2$*, and* $n_a\, ef + n_b(f^2 - e^2) + n_c f^2 = e^2$ *has* no *solution in nonnegative integers, for any* $1 \leq e < f$ *with* $\gcd(e, f) = 1$*;*

(iii) *hence some far-side edge crosses the far endpoint of T_2's c-edge* straight through*: every such tiling contains a* ***pierced*** β***-corner*** *at distance f^2 along an inner apex ray, and the angles on T_2's side of the piercing line beyond the corner form $\{\alpha, \gamma\}$ or $\{3\alpha, \beta\}$.*

Proof. (i) The apex is 3α (three tiles, each α there); the two edges of an α-corner are b and c. The boundary edges at the apex are the sides' last edges, c by hypothesis, so the outer tiles point b inward and T_2 has $\{b, c\}$ inward. Whichever inner ray pairs T_2's c with an outer b, the b-edge ends at distance b: its far endpoint is the γ-corner of its tile (b joins α to γ), and the vertex figure on that side of the straight c-edge sums to π with a γ present, hence is $(1, 1, 1)$ by the vertex classification. (ii) Any solution has $n_a = n_c = 0$ since $ef, f^2 > e^2$; then $n_b(f^2 - e^2) = e^2$ gives $n_b f^2 = (n_b + 1)e^2$, and $\gcd(e, f) = 1$ forces $f^2 \mid n_b + 1$, so $n_b \geq f^2 - 1$ and $e^2 = n_b(f^2 - e^2) \geq f^2 - 1 \geq e^2$, with equality forcing $f^2 - e^2 = 1$ — impossible, as $f^2 - e^2 \geq 2e + 1 \geq 3$. (iii) The far side of the leftover segment is covered by consecutive whole tile edges starting at the T-junction; by (ii) no partial sum lands exactly on the segment's far endpoint, so some edge strictly crosses it. That endpoint is the β-corner of T_2 (c joins α to β). The angles on T_2's side of the crossing line beyond the corner fill $\pi - \beta$; writing them as $x\alpha + y\beta + z\gamma$ and clearing β, γ, irrationality of α/π forces $y + z = 1$, $2x + z = 3y + 3$, whose only solutions are $(3, 1, 0)$ and $(1, 0, 1)$. □

Remark 14 (Validation on the 44-tiling)**.** The theorem can be checked against the genuine 44-tiling ($m = 2$; its equal sides do end with c), where it holds exactly: the apex rays pair one $b \,|\, b$ and one mismatch $c \,|\, b$; the middle tile's c-endpoint $V = (10, 2\sqrt{15})$ is pierced by a straight b-edge, and the sector on the middle tile's side is $\beta + \{\alpha, \gamma\}$ — the first continuation type. Parts (ii) and the vertex figures in (i), (iii) are machine-checked (`apex_leftover_nonrepresentable`, `pierced_corner_types`, `BaseBetaWalks.lean`). The configuration is an interior constraint, uniform in (e, f) and unconditional at $m = 1$: every hypothetical thick-regime tiling, $N = 59$ included, must contain it.

Proposition 15 (Unsplittability, and the rigidity of the thick regime)**.** *For every coprime $1 \leq e < f$, neither $a = ef$ nor $b = f^2 - e^2$ is a sum of two or more tile edges; and $c = f^2$ is such a sum if and only if $e = 1$, in which case uniquely $c = a^f$. Consequently* in the entire open thick regime, where $e \geq 2$, no tile edge splits at all.

Proof. Reduce $n_a\, ef + n_b(f^2 - e^2) + n_c f^2 = L$ modulo f. For $L = a$: $f \mid n_b e^2$, so $f \mid n_b$ by $\gcd(e, f) = 1$; $n_b \geq f$ overshoots since $fB > ef$ (as $B = f^2 - e^2 \geq 2e + 1 > e$), so $n_b = 0$, and cancelling f leaves $n_a e + n_c f = e$, forcing $n_c = 0$ and $n_a = 1$. For $L = b$: $f \mid (n_b - 1)e^2$, so $n_b \equiv 1$; $n_b \geq 1 + f$ overshoots, so $n_b = 1$ and the rest vanishes, giving $n_a = n_c = 0$. Either way the total is 1, never ≥ 2. For $L = c$: $f \mid n_b$ forces $n_b = 0$ as before, and $n_a e = f(1 - n_c)$ gives $n_c = 0$ and $e \mid f$, i.e. $e = 1$, $n_a = f$. □

Remark 16 (The golden ratio)**.** Two consequences. First, Proposition 15 is exactly the hypothesis the corner parallelogram needs, so Remark 45(b) holds with *no* proviso. Second, $e = 1$ — which both genuine tilings realize — is the only case in which an edge splits at all, while the open $e \geq 2$ regime is rigid. The comparison of a with b must be made with care: $b > a \iff f^2 - ef - e^2 > 0 \iff f/e > \varphi$, the golden ratio. The thick regime straddles that threshold: $N = 66$ $(3, 5)$ and $N = 131$ $(4, 7)$ are super-golden, while $N = 23$ $(2, 3)$, $N = 59$ $(4, 5)$, $N = 83$ $(5, 6)$ and $N = 167$ $(5, 8)$ have $b < a$. Proposition 15 above assumes no inequality between a and b.

Proposition 17 (Rung two: the pre-piercer chain)**.** *In the configuration of Theorem 13, every full far-side edge strictly between the T-junction and the piercer lies inside $[b, f^2]$, hence has length $\leq e^2$; among the tile edges only $b = f^2 - e^2$ can satisfy this, and only in the* super-thick *regime $f^2 \le 2e^2$. So the piercer starts at $(t+1)b$ with $tb < e^2$, where $t = 0$ outside super-thick; and if the piercer is itself a b-edge, its corner at its starting junction is exactly α (the $(1,1,1)$ figure offers $\{\alpha, \beta\}$, while a b-edge's endpoints are $\{\alpha, \gamma\}$). In particular for $N = 59$ ($(e, f) = (4, 5)$, super-thick: $25 \le 32$) the piercer starts at distance 9 or 18, and for every non-super-thick member it starts at the T-junction itself.*

Remark 18 (Validation on the 44-tiling)**.** On the genuine 44-tiling ($f^2 = 4 > 2e^2 = 2$, so $t = 0$) the piercer spans exactly $[3, 6]$, starting at the T-junction, with corner α there to numerical precision, as forced; the arithmetic core is machine-checked (`pre_pierce_dichotomy`). The rung also has a computational consequence: the apex neighbourhood is rigid (three forced α-tiles, a forced mismatch ray, a forced T-junction, a forced piercer start), while the base admits several walks. Since the corner-anchored search anchors at the lowest vertex, orienting the target apex-down (a reflection, under which verdicts are preserved) places the rigid chain at the root of the search tree: on $N = 23$ the apex-down search exhausts in 3,954 nodes against 19,677, whereas on the thin member $N = 47$, where the base is the rigid side, the apex-down orientation is slower. The thick searches $N = 59, 66$ are run apex-down.

Theorem 19 (Alignment of the mismatch ray)**.** *Let $m = 1$, $e \ge 2$, and let r be the mismatch inner apex ray of Theorem 13, of inner length $f \cdot b$ (Proposition 17, $b = f^2 - e^2$). Along r, the near side is $T2$'s single unsplittable c-edge covering $[0, f^2]$, and the far side begins with $T1$'s b-edge. Then:*

(i) no common junction *lies in $(0, f^2)$: for $1 \le k \le f - 1$, $k \cdot b$ is never $n_a\, ef + n_b\, b + n_c f^2$ with $n_c \ge 1$ (so every far-side junction is a genuine T-junction, the near c-edge passing straight through it);*

(ii) *the far side is exactly b^f: the only representation of $f \cdot b$ as $n_a\, ef + n_b\, b + n_c f^2$ with $n_b \ge 1$ is $(0, f, 0)$;*

(iii) *$V = f^2$ is pierced: $b \nmid f^2$, so no far-side junction $k \cdot b$ equals f^2 and a far b-edge straddles $T2$'s β-corner.*

*All three are machine-checked (`far_near_disjoint`, `far_is_bpow`, `b_not_dvd_fsq`); (ii) alone uses γ-vertex induction (*via *the walk arithmetic) to fix the first far edge, the rest is pure number theory.*

Proof. (i) If $kb = n_a ef + n_b b + n_c f^2$ then $(k - n_b)b = f(n_a e + n_c f)$, so $f \mid (k - n_b)e^2$ and $\gcd(e, f) = 1$ give $f \mid k - n_b$; writing $k - n_b = fs$, the equation reads $sb = n_a e + n_c f \ge f > 0$ (as $n_c \ge 1$), so $s \ge 1$ and $k \ge n_b + f \ge f$, against $k \le f - 1$. (ii) $f \mid n_b$ by the same reduction, and $n_b \le f$ by size, so $n_b \in \{0, f\}$; $n_b \ge 1$ forces $n_b = f$, and then $n_a ef + n_c f^2 = 0$ gives $n_a = n_c = 0$. (iii) $b \mid f^2$ would give $b \mid e^2$, hence $b \mid \gcd(f^2, e^2) = 1$, so $b = 1$, impossible as $b \ge 2e + 1 \ge 5$. The far junctions are the multiples kb, none equal to f^2, so the b-edge containing f^2 straddles it. □

Theorem 20 (The mismatch ray is completely determined)**.** *Let $m = 1$, $e \ge 2$, and let the family be thick, $f \le 2e$. Then along the mismatch inner apex ray, of inner length $f\, b$:*

$$\textit{far side} = b^f \ \ (f \ \textit{edges}), \qquad \textit{near side} = \{a^{f-e},\, c^{f-e}\}\ (2(f-e)\ \textit{edges}),$$

and the near side is the unique *possibility. In particular the near side carries equally many a- and c-edges, and no b-edge.*

Proof. The far side is Theorem 19(ii). For the near side, $T2$'s c-edge covers $[0, f^2]$ and the remainder $f\, b - f^2 = f(b - f)$ is covered by whole edges, say n_a of length a, n_b of length b and n_c of length c. Reducing modulo f kills n_b as in Theorem 19(i), and cancelling one factor f leaves

$$n_a e + n_c f + f = b.$$

Now $(n_a, n_c) = (f-e,\ f-e-1)$ solves it, since $(f-e)e+(f-e-1)f = f^2-e^2-f = b-f$. For uniqueness, $\gcd(e,f)=1$ and the equation give $f \mid n_a+e$, so $n_a+e = ft$; $t=0$ is impossible and $t \ge 2$ would force $n_a e \ge (2f-e)e$, whence $2ef \le f^2-f$, i.e. $f \ge 2e+1$, contradicting thickness. Hence $t=1$, $n_a = f-e$, and then $n_c f = f(f-1-e)$ gives $n_c = f-e-1$. Adding $T2$'s own c-edge produces the multiset $\{a^{f-e}, c^{f-e}\}$, whose total length is $(f-e)(ef+f^2) = f(f-e)(e+f) = f\,b$ as required. □

Remark 21 (A parity argument that fails, and why)**.** It is tempting to argue as follows. Every tile has exactly one b-edge; by Proposition 15 the edge b is never a sum of two or more tile edges; so an interior b-edge is matched by exactly one further b-edge, interior b-edges pair the tiles two at a time, and $N \equiv Q_\partial \pmod 2$ with Q_∂ the number of boundary b-edges. We record that this is *false*, because the middle step does not follow.

Unsplittability forbids an edge from being *exactly partitioned* by two or more tile edges. It does not forbid a *straddle*: a longer edge may contain a b-edge in its interior and overhang it at one or both ends, and then the b-edge has no partner. Indeed the apex-mismatch configuration of Theorem 13 is built on precisely such a straddle. The genuine tilings confirm the failure: in the 44-tiling six b-edges belong to a single tile although *no* b-edge lies on the boundary, and the 99-tiling has thirty-three matched pairs, eleven boundary b-edges and twenty-two interior straddles. The correct relation is

$$N = 2 \cdot \#\{\text{matched pairs}\} + \#\{\text{boundary}\} + \#\{\text{straddled}\},$$

so the parity conclusion needs the straddle count to be even, which nothing supplies. (It happens to be even, 6 and 22, in these two tilings, which is why the conclusion survives there.)

The same objection defeats the analogous arguments for the a- and c-edges. We state this explicitly because the argument is natural, its conclusion is correct on every example we can test, and it is nevertheless not a proof.

Theorem 22 (Unique representation of $k\,b$)**.** *For $1 \le k \le f-1$ the only way to write $k\,b$ as $n_a\,ef + n_b\,b + n_c f^2$ in nonnegative integers is b^k. (Machine-checked, `kb_unique_rep`; it generalizes Theorem 19*(ii)*, which needed $n_b \ge 1$.)*

Theorem 23 (The angle $(\pi-\alpha)/2$)**.** *A vertex figure filling $(\pi-\alpha)/2$ is $\alpha+\beta$, and nothing else. (Machine-checked, `halfpi_minus_alpha_unique`.)*

Theorem 24 (The b-run orientation lemma)**.** *Let a run of b-edges be laid end to end along a ray, seeded at an apex tile. Then every b-edge of the run carries α at its apex end and γ at its far end, and every junction interior to the run is an (α,β,γ) π-vertex. In particular the $3\alpha+2\beta$ figure never occurs along such a run.*

Proof. A b-edge joins its tile's α- and γ-corners, so at a junction the incoming tile presents α or γ, and likewise the outgoing one. The $3\alpha+2\beta$ figure contains no γ, and two γ's cannot share a side of a straight line since $2\gamma = \pi+\alpha > \pi$. Hence if the incoming corner is γ the figure is $(1,1,1)$, whose single α must be the outgoing corner, so the next edge again ends in γ. An apex tile has α at the apex, hence γ at the far end of its b-edge, which seeds the induction; it proceeds with no branching. □

Theorem 25 (The far region is a scaled tile)**.** *Suppose the mismatch apex ray is a complete wall, of inner length $f\,b$. Then it cuts off a triangle with angle α at the apex, β at the base corner and hence γ at its third vertex, and with sides $f\,c$, $f\,a$, $f\,b$ — that is,* the tile scaled by f. *It therefore has area f^2 times the tile's and is tiled by the standard f-subdivision into f^2 copies, whose ray side is exactly b^f with the junction chain of Theorem 24.*

Corollary 26 (The far side can never give a contradiction)**.** *Under the hypothesis of Theorem 25 — that the mismatch apex ray is a complete wall — no obstruction to a base-β tiling can be derived from the far side of that ray, at any (e, f): that side admits a genuine tiling. The hypothesis may not be dropped; if the ray is not a complete wall the far region is not a scaled copy of the tile and the conclusion does not follow.*

Remark 27 (Where the obstruction must instead live)**.** Corollary 26 is a negative result and an important one: it explains why the pierced-corner analysis resisted three independent attempts. The configuration it pins down is not contradictory but *realizable*, so no amount of further chasing on that side can close the branch. The check is exact for every family — for $(e, f) = (4, 5)$ the far region has sides $(45, 100, 125) = 5 \cdot (9, 20, 25)$ and contains $25 = f^2$ of the 59 tiles — and it is confirmed on the genuine 44-tiling, whose mismatch ray has *no* straddling tile and whose far region contains $16 = (mf)^2$ tiles with sides $4 \cdot (c, a, b)$.

The live direction is the other ray. For every thick member with $m = 1$, $e \geq 2$, the *matched* inner apex ray is *not* a wall. Indeed if it were, the target would split into $f^2 + f^2 + (f^2 - e^2)$ tiles, the last a middle triangle M with apex α, legs $f\,b$ and base $e\,b$. Since $k\,b$ has the unique representation b^k for $1 \leq k \leq f - 1$ (Theorem 22), both edges of M at its far corner are b-edges; but the angle there is $(\pi - \alpha)/2$, whose only decomposition is $\alpha + \beta$ (Theorem 23), and a b-edge endpoint is an α- or γ-corner, so no tile present can supply the β. This is the precise mechanism behind the failure of the clean-ray corollary retracted in Remark 28, and it is confirmed on the 44-tiling, whose matched ray carries exactly three straddling tiles against the mismatch ray's zero.

Consequently the branch now reduces to a single question: *can the residual region — the target minus the far wall, containing $N - f^2$ tiles and crossed by a necessarily straddled matched ray — be tiled?* Every tool developed here (walk arithmetic, vertex figures, unsplittability) is wall-based and goes silent once a tile straddles. What is needed is a bound on where a tile may straddle a ray whose two ends are pinned by b-edges.

Remark 28 (Two corrections this forces)**.** Theorem 24 is uniform in (e, f) and in m, and is confirmed on the genuine 44-tiling, where all four far b-edges of the mismatch ray and both sides of the clean ray read $\alpha \to \gamma$. It strictly generalizes Proposition 17, which established only the first step. But it also refutes two statements of an earlier version, which we withdraw.

First, the assertion that the *clean* apex ray is b^f against b^f. Its justification — "by the symmetry of the isosceles target both inner rays have the same inner length $f\,b$" — conflates two objects. The symmetry does apply to the geometric rays; it does not apply to the tiling's maximal edge *segment* along a ray, which may terminate early at a vertex where the edges turn away. In fact the clean ray's segment can never reach the base: if it did, both final b-edges would contribute γ at an interior point of the base, where the tiles above sum to π, contradicting $2\gamma > \pi$. On the 44-tiling the two segments have lengths 12 and 9 respectively, and the clean ray terminates at an interior $\beta + 3\gamma$ vertex — which is precisely the M_4 vertex the census forces.

Second, and more seriously, the same objection applies to the hypothesis under which Theorems 19 and 20 were stated: that the *mismatch* ray's maximal segment attains the full length $f\,b$. That too was justified by symmetry, and it too now requires a genuine proof. The arithmetic of those theorems is unaffected — given a chain of total length $f\,b$ containing a b-edge, the conclusions stand and remain machine-checked — but their geometric hypothesis must be stated explicitly. Establishing that the mismatch segment reaches the base is the single highest-value open step in this branch: it would restore Theorems 19 and 20 unconditionally for every thick member.

Remark 29 (Scope and consequences)**.** Thickness is used exactly once, and sharply: the next solution would be $n_a = 2f - e$, which needs $f \geq 2e + 1$ — the thin regime. So in the regime that is open, the entire mismatch ray is now pinned on *both* sides, with no freedom beyond the order of the near-side edges. Explicitly: $N = 23$ has far 5^3, near $\{6, 9\}$; $N = 59$ far 9^5, near $\{20, 25\}$; $N = 66$ far 16^5, near $\{15^2, 25^2\}$; $N = 83$ far 11^6, near $\{30, 36\}$; $N = 167$ far 39^8, near $\{40^3, 64^3\}$. The count of interior junctions is therefore also determined — $f - 1$ on the far side and $2(f - e) - 1$ on the near side, none shared (Theorem 19(i)) — so a hypothetical thick tiling carries exactly $3f - 2e - 2$ T-junctions along this ray, each a π-vertex. The arithmetic is machine-checked (`near_side_unique`, `BaseBetaWalks.lean`).

Remark 30 (What the alignment pins, uniformly)**.** For every open thick member the mismatch ray is now completely determined out to length $f \cdot b$: its far wall is b^f, terminating transverse to the opposite side, and $V = f^2$ falls strictly inside the $\lceil f^2/b \rceil$-th b-edge. Explicitly (all exact): $N = 23$ far side $b^3 = 5^3$, $V = 9 \in (5, 10)$; $N = 59$ $b^5 = 9^5$, $V = 25 \in (18, 27)$; $N = 66$ $b^5 = 16^5$, $V = 25 \in (16, 32)$; $N = 83$ $b^6 = 11^6$, $V = 36 \in (33, 44)$. This pins an extended *interior* structure of a hypothetical thick tiling, uniformly in (e, f); it serves as a root prune for the apex-down search; the forced b^f wall must land on the base at coordinate $e \cdot f^2$, whose b-count is bounded (`base_b_bound`). It does not, on its own, close any member.

Remark 31 (Scope, and why $m = 1$ is the rigid case)**.** Theorem 12(i) generalizes the $e = 1$ base analysis used for Theorem 10 to the infinite family $f > 2e$ — $e = 1$, $f \geq 3$; $e = 2$, $f \geq 5$; $e = 4$, $f \geq 9$; $\ldots$ — covering the primes $N = 47, 71, 107, 191, 227, 239, 359, \ldots$, and its parts (iii)–(iv), which need no thinness, deliver the $e \geq 2$ analogue of the $e = 1$ structure theorem of Remark 11 for *every* (e, f): the equal sides are b-free and the base's b-count is bounded by $j(f - e) \leq e - 1$. The step that fails for $m \geq 2$ is the very first one: there $Q \equiv em \pmod f$, so j may be *negative* and the level rises to $2em$. This is not an artifact. The genuine 44-tiling ($e{=}1, f{=}2, m{=}2$) has base walk *aaaaccca*, i.e. $(P, Q, R) = (5, 0, 3)$, which is the parameter $j = -1$, $p = 3$; and the 99-tiling ($e{=}1, f{=}2, m{=}3$) has base walk *aabbbbbbbcc*, i.e. $(2, 7, 2)$, the parameter $j = 2$, $p = 0$. Both satisfy $R \geq 1$, as the γ-trap requires.

Remark 32 (Why the colouring method cannot close the rest)**.** The residual gap ($e \geq 2$) is not merely unproved by [2, Thm. 14]: it is *unprovable by that method*. Let $\eta = e^{i(\omega + \pi/2)}$, so that $f\eta^2 + e\eta + f = 0$ and every edge direction of every such tiling lies in $\langle \eta, -1 \rangle$. Any T-junction-proof invariant of the shape $\sum_{\text{edges}} L \cdot \chi(\theta)$ with $\chi(-w) = -\chi(w)$ — i.e. exactly the Φ/colouring method, and exactly Beeson's — is equivalent to the master identity

$$m\big(-c\, z^4 + a\, z^3 + b\, z^2 + a\, z - c\big) = z^3 (f - ez) P(z) + (fz - e) Q(z), \qquad P, Q \in \mathbb{Z}[z, z^{-1}],$$

whose $z = \pm 1$ specializations return exactly $M = m(f + e)$ and $-m(f - e)$. But that identity is *solvable for every* m, with the explicit witness $P_0 = mf(z^{-1} - z^{-3})$, $Q_0 = m(ez^2 - fz^3)$. Hence no invariant in this class can force $g \mid M$, for any (e, f, m): the Φ-method — the engine of both Beeson's argument and of our own $2\pi/3$ theorem — provably cannot decide the base-β residue. Closing it requires genuinely non-linear input.

Remark 33 (Beeson's Theorem 14 is *false*, not merely unsound)**.** The exhaustive engine settles the status of $g \mid M$ outright. The isosceles triangle $(24, 24, 33)$ with base angles β *is* tiled by 99 copies of the tile $(2, 3, 4)$: this is the base-β family at $(e, f, m) = (1, 2, 3)$, so $X = f^3 m = 24$, $Y = em(3f^2 - e^2) = 33$, $N = m^2(3f^2 - e^2) = 99$ and $M = (f + e)m = 9$. The tiling was found by exact search and re-verified independently (congruence of all 99 tiles, disjointness, containment, exact area) against a separately constructed instance. Its boundary base walk is the multiset $\{a^2, b^7, c^2\}$ $(2 \cdot 2 + 7 \cdot 3 + 2 \cdot 4 = 33)$, consistent with $q_{\text{base}} \equiv em \equiv 1 \pmod 2$.

But $g = \gcd(a, c) = \gcd(2, 4) = 2$ and $M = 9$, so $g \nmid M$. Hence

[2, Thm. 14]**'s conclusion** $g \mid M$ **is FALSE,** not merely unsoundly proved (Remark 9); equivalently $f \mid m$ fails, here $2 \nmid 3$. Three consequences. (i) The base-β prime exclusion cannot rest on Theorem 14 *at all*; only a per-value search remains, which is what we do. (ii) Our own base-$(\alpha+\beta)$ and base-α replacements (Propositions 47, 8) are unaffected: both were deliberately built to use *no* $g \mid M$. (iii) The values whose exclusion had rested only on $g \mid M$ $(21, 39, 66, 70)$ are demoted (Remarks 79, 9); 21 and 39 are now excluded by exhaustive search instead. Finally $N = 99$ is itself a new realization: 99 is not a square, a sum of two squares, nor $2n^2, 3n^2, 6n^2$, so it is not commensurably realizable.

Proposition 34 (Dissection basics)**.** *Let ABC be cut into N triangles $T_1, \dots, T_N$ with pairwise disjoint interiors and union ABC. Then $\sum_i |T_i| = |ABC|$, and if the T_i are pairwise congruent to a tile T then $|ABC| = N|T|$. Moreover, at a point $v \in ABC$ the tiles whose closure contains v subtend angles summing to 2π if v is interior to ABC, to π if v is interior to a side, and to the corner angle if v is a corner; each summand is a tile angle α, β, γ if v is a vertex of that tile, and π if v lies in the relative interior of one of its edges. Hence the sum has the form $p\alpha + q\beta + r\gamma + s\pi$ with $p, q, r, s \geq 0$ integers.*

Proof. The area identity is machine-checked in `lean/Dissection.lean` (`Dissection.volume_target`, `volume_target_of_congruent`), in `EuclideanSpace` $\mathbb{R}^2$ with Lebesgue measure and axiom-clean: the tiles are convex with positive volume and null frontier (`Convex.addHaar_frontier`), so the union is almost-everywhere disjoint and volumes add. The angle statement is the local structure of a dissection: a small disc about v meets finitely many tiles, each in a single circular sector, and the sectors tile the disc's intersection with ABC. The $s\pi$ term is the contribution of tiles meeting v in an edge interior; it is present exactly when the dissection is not edge-to-edge at v, which does occur here (Remark 21). Mathlib has no theory of planar dissections, so this angle statement is not formalized; it is the geometric input flagged in Section 8. What *is* formalized is the arithmetic consequence drawn from it in Proposition 35 (`vertex_multiplicities` and its three corollaries) and the angle sum of a tile (`cornerAngle_sum`). □

Proposition 35 (Vertex figures)**.** *Let the tile satisfy $3\alpha + 2\beta = \pi$ with $\alpha \notin \mathbb{Q}\pi$, so $\gamma = 2\alpha + \beta$ and $2\gamma = \pi + \alpha$. If the angles at v have multiplicities (p, q, r) for (α, β, γ) together with s straight angles, then $p\alpha + q\beta + r\gamma = t\pi$ with $t = 1$ or $t = 2$, and*

$$t = 1: \ (p, q, r) \in \{(3, 2, 0), \ (1, 1, 1)\}; \qquad t = 2: \ (p, q, r) \in \{(6, 4, 0), \ (4, 3, 1), \ (2, 2, 2), \ (0, 1, 3)\}.$$

A junction interior to a side of ABC has $t = 1$; an interior vertex has $t = 2 - s$, so $t \in \{1, 2\}$. In particular, in every case $r \leq 1$ when $t = 1$.

Proof. Write $\beta = (\pi - 3\alpha)/2$ and $\gamma = (\pi + \alpha)/2$; then $p\alpha + q\beta + r\gamma = \frac{1}{2}\big((2p - 3q + r)\alpha + (q + r)\pi\big)$. By Proposition 34 this equals $(t - s')\pi$ for integers, and since α/π is irrational (`tile_alpha_irrational`) both $2p - 3q + r = 0$ and $q + r = 2t$ follow. This is exactly `vertex_multiplicities`, machine-checked. Enumerating the non-negative solutions of $q + r = 2t$, $2p = 3q - r$ gives the two lists (`vertex_pi`, `vertex_beta_corner`, `vertex_apex` in `BaseBetaE1.lean`); that a geometric vertex figure supplies those integer equations is `vertex_pi_multiplicities`, `vertex_beta_corner_multiplicities` and `vertex_apex_multiplicities` in `Dissection.lean`. □

Proposition 36 (Corner figures)**.** *On a base-β target, the vertex figure at each base corner is a* single *tile presenting β, and the figure at the apex is* exactly three *tiles, each presenting α. In particular no corner of ABC carries a γ. The two edges of the base-corner tile at that corner are a and c; the edges of the apex tiles at the apex are b and c.*

Proof. By Proposition 35, an angle relation $p\alpha + q\beta + r\gamma = t$ at a corner splits into $2p - 3q + r = \sigma$ and $q + r = \tau$ for integers determined by t. A base corner has $t = \beta$, giving $\sigma = -3$, $\tau = 1$; the case $(q,r) = (0,1)$ would need $2p = -4$, so $(p,q,r) = (0,1,0)$. The apex has $t = 3\alpha$, giving $\sigma = 6$, $\tau = 0$, hence $q = r = 0$ and $p = 3$. The edge assertions are the incidence rule: a joins the β,γ corners, b joins α,γ, c joins α,β. Machine-checked (`corner_beta_unique`, `corner_apex_unique` in `lean/BaseBetaCorners.lean`). □

Proposition 37 (γ-trap)**.** *Every side of a base-β target carries at least one c-edge.*

Proof. A junction interior to a side is a π-vertex, so by Proposition 35 its multiplicity vector is $(3,2,0)$ or $(1,1,1)$: it carries *at most one* γ (`pi_vertex_gamma_le_one`).

Let a side of ABC be partitioned by the tile edges $E_1,\dots,E_n$ lying on it, with junctions $J_0,\dots,J_n$, where J_0 and J_n are corners of ABC. (The partition is exact: the tiles cover ABC with disjoint interiors, so each point of the side lies in exactly one tile edge. This is a statement about the boundary and is unaffected by non-edge-to-edge incidence in the interior.)

Suppose the side carries no c-edge, so every E_i is an a- or a b-edge. Both are incident to γ at exactly one endpoint, so each E_i places a γ at J_{i-1} or at J_i, and by the previous paragraph no junction receives two. By Proposition 36 neither J_0 nor J_n carries a γ at all. Hence E_1 must place its γ at J_1; that junction is then full, so E_2 must place at J_2; inductively E_i places at J_i, and finally E_n places at J_n — a corner. Contradiction. □

Remark 38. The hypothesis that the corner angles are $\beta,\beta,3\alpha$ is essential, and this is why [2, Lem. 6] is false as quoted (Remark 5): on the tile-similar target a corner *is* a γ, the chain terminates harmlessly, and the counterexample there has two sides with no c-edge. On a base-β target no corner is a γ, and the argument runs.

Proposition 39 (b is unsplittable)**.** *b is not a sum of two or more tile edges: if $n_a\,a+n_b\,b+n_c\,c = b$ with $n_a,n_b,n_c \ge 0$, then $(n_a,n_b,n_c) = (0,1,0)$.*

Proof. If $n_b = 0$ the remaining terms are divisible by f, so $f \mid b$; with $b + e^2 = f^2$ this gives $f \mid e^2$, and $\gcd(e,f) = 1$ forces $f = 1$, contradicting $1 \le e < f$. If $n_b \ge 2$ then $n_b b > b$ already, as $b > 0$. So $n_b = 1$, the remaining terms sum to zero, and $a,c > 0$ give $n_a = n_c = 0$. Machine-checked (`b_unsplittable`). □

Proposition 40 (Corner parallelogram)**.** *Let T_A be the tile at a base corner A. Its b-edge is a chord $[P,Q]$ with both endpoints on ∂ABC, matched by exactly one tile T_3, which presents α where T_A presents γ and conversely; so $T_A \cup T_3$ is a parallelogram with sides a and c. Consequently the first* two *edges of each of the two sides meeting at A lie in $\{a,c\}$.*

Proof. By Proposition 36, T_A is the unique tile at A and its two edges there are a and c, one along each side of ABC; so its other two vertices P (the γ corner) and Q (the α corner) lie one on each of those two sides. Its third edge is the chord $[P,Q]$.

The tiles on the far side of $[P,Q]$ have edges covering it. No such edge can straddle: a straddling edge would extend beyond P or beyond Q along the line PQ, and both P and Q lie on ∂ABC, so the extension leaves ABC. Hence the far-side edges partition $[P,Q]$ exactly, and by Proposition 39 the partition is trivial: a single tile T_3 with b-edge $[P,Q]$.

T_A presents γ at P and α at Q. Both P and Q are interior to sides of ABC, hence π-vertices, which carry at most one γ; since T_3's b-edge is incident to α and γ, it must present α at P and γ at Q. Two congruent triangles glued along b with α,γ interchanged form a parallelogram with sides a,c.

For the last assertion, take the side on which Q lies and let E_1 be its first edge (an edge of T_A). At Q the angles are T_A's α, T_3's γ, and, since $\alpha+\gamma = \pi/2+3\theta$, a remaining $\beta = \pi/2-3\theta$ carried by one further tile. That tile owns E_2 and presents β, whose incident edges are a and c. The same argument at P handles the other side. □

Theorem 41 (Walk structure at $m = 1$)**.** *Let $m = 1$, $\gcd(e, f) = 1$, $1 \le e < f$, and consider a tiling of the base-β target.*

(i) *If $f^2 > 2e^2$, then each equal side carries* no b-edge. *Its edges are a's and c's with $n_a = fk$ and $n_c = f - ke \ge 1$, and it ends with a c-edge at the apex.*

(ii) *If $f^2 > 2ef + e^2$ — equivalently $f/e > 1 + \sqrt{2}$ — then the base carries* exactly e b-edges, *and $n_a = f\ell$, $n_c = e(2 - \ell)$ with $\ell \in \{0, 1\}$. In particular*

$$R_{\text{base}} \in \{e,\, 2e\}.$$

Proof. A side of length L is exactly partitioned by the tile edges on it, giving $n_a\, ef + n_b\, b + n_c\, f^2 = L$ with $n_a, n_b, n_c \ge 0$, and $n_c \ge 1$ by Proposition 37.

(i) Here $L = f^3$. Rearranging with $b = f^2 - e^2$ gives $n_b e^2 = f\,(n_a e + n_b f + n_c f - f^2)$, so $f \mid n_b e^2$ and hence $f \mid n_b$ by coprimality; write $n_b = ft$. If $t \ge 2$ the b-edges alone cost at least $2fb = 2f^3 - 2fe^2$, which together with $n_c f^2 \ge f^2$ exceeds f^3 unless $f^2 \le 2e^2$. If $t = 1$, cancelling f leaves $n_a e + n_c f = e^2$, so $n_a \le e$ and $f \mid e(e - n_a)$; coprimality with $0 \le e - n_a \le e < f$ forces $n_a = e$ and then $n_c = 0$, contradicting $n_c \ge 1$. So $t = 0$. With $n_b = 0$ the equation is $n_a e + n_c f = f^2$, whence $f \mid n_a$; writing $n_a = fk$ gives $n_c = f - ke$. Finally, the apex tile on that side presents α (Proposition 36), whose incident edges are b and c; as the side has no b-edge, it ends with a c-edge.

(ii) Here $L = e(3f^2 - e^2)$, and the same rearrangement gives $(n_b - e)e^2 = f\,(n_a e + n_b f + n_c f - 3ef)$, so $f \mid n_b - e$; write $n_b = e + ft$. Negative t makes $n_b < 0$. For $t \ge 1$ the b-edges alone cost at least

$$(e + f)b = (f + e)^2(f - e) = e(3f^2 - e^2) + f\,(f^2 - 2ef - e^2),$$

which already exceeds L under the hypothesis, before the mandatory c-edge is placed. So $n_b = e$; subtracting and cancelling f leaves $n_a e + n_c f = 2ef$, whence $f \mid n_a$, and writing $n_a = f\ell$ gives $n_c = e(2 - \ell)$. Then $n_c \ge 1$ forces $\ell \le 1$.

Machine-checked (`equal_side_no_b`, `equal_side_shape`, `base_b_count` in `lean/BaseBetaWalkArith.lean`), and cross-checked by enumerating all walk solutions for the 198 coprime pairs with $f \le 25$: no discrepancies. ☐

Remark 42. Both hypotheses are sharp: $n_b = 2f$ on an equal side is admissible exactly when $f^2 \le 2e^2$, and $n_b = e + f$ on the base exactly when $f^2 \le 2ef + e^2$, by the displayed identity. Of the 42 base-β prime candidates below 1000, 28 satisfy (i) — including the thick values 23, 131 and 167, since (i) needs only $f/e > \sqrt{2}$ — and 17 satisfy (ii). The theorem structures those cases; it does not by itself exclude any of them.

Theorem 43 (The $e = 1$ subfamily reduces to one walk)**.** *Let $e = 1$, $m = 1$ and $f \ge 3$, so the target is $(f^3, f^3, 3f^2 - 1)$, the tile is $(f, f^2 - 1, f^2)$ and $N = 3f^2 - 1$. Both hypotheses of Theorem 41 hold ($f^2 > 2$ and $f^2 > 2f + 1$). In any tiling:*

(i) *neither equal side carries a b-edge; each equal side begins and ends with a c-edge, and therefore carries at least two;*

(ii) *the base carries exactly one b-edge and exactly one c-edge — so $R_{\text{base}} = 1$ — and its first and last edges are a-edges.*

Thus the base walk is a permutation of (a^f, b, c) beginning and ending with a, with the b in position $3, \ldots, f$.

Proof. *(i)* An equal side has length f^3 and its edges lie in $\{f,\ f^2 - 1,\ f^2\}$. Reducing $n_a f + n_b(f^2 - 1) + n_c f^2 = f^3$ modulo f gives $f \mid n_b$. If $n_b \ge 2f$ then $n_b(f^2 - 1) \ge 2f^3 - 2f > f^3$ for $f \ge 2$. If $n_b = f$ then $f(f^2 - 1) = f^3 - f$ leaves f, forcing $n_c = 0$ and $n_a = 1$, hence no c-edge on that side — contradicting Proposition 37. So $n_b = 0$, and then $n_a + n_c f = f^2$, so $f \mid n_a$; write $n_a = fk$, $n_c = f - k$.

By Proposition 36 the apex tile on that side presents α, whose incident edges are b and c; since the side carries no b-edge, it ends with a c-edge. Its other end is handled in *(ii)*: there we show the base begins and ends with a-edges, so the base-corner tile's c-edge lies on the equal side, which therefore begins with a c-edge. Were $n_c = 1$ the single c-edge would be both the first and the last, forcing the side to consist of one edge and $f^3 = f^2$; so $n_c \geq 2$.

(ii) The base has length $3f^2 - 1$. Reducing modulo f gives $n_b \equiv 1$. If $n_b \geq f + 1$ then $n_b(f^2 - 1) \geq f^3 + f^2 - f - 1 > 3f^2 - 1$ for $f \geq 3$, so $n_b = 1$. Subtracting, $2f^2 = n_a f + n_c f^2$, so $n_a = f(2 - n_c)$ with $n_c \in \{0, 1, 2\}$, and $n_c \geq 1$ by Proposition 37.

If $n_c = 2$ then $n_a = 0$ and the base is three edges, a permutation of (b, c, c). By Proposition 40 the first two edges lie in $\{a, c\}$ and so do the last two; with no a-edge available this forces all three to be c-edges, contradicting $n_b = 1$. Hence $n_c = 1$ and $n_a = f$, so the base is a permutation of (a^f, b, c) with $n = f + 2$ edges, and Proposition 40 places the b away from the first two and last two positions.

It remains to exclude $E_1 = c$. Suppose it. Then the unique c-edge is E_1, so every one of $E_2, \ldots, E_n$ is an a- or b-edge and places a γ at one of its endpoints; there are $f + 1$ of them and $f + 1$ interior junctions $J_1, \ldots, J_{n-1}$. The corner tile T_A has its c-edge on the base, so its α vertex is J_1 and its γ vertex lies on the equal side; by Proposition 40 its partner T_3 presents γ at J_1. So J_1 already carries a γ, and since a π-vertex carries at most one (Proposition 35), E_2 must place its γ at J_2; inductively E_i places at J_i, and E_n places at J_n, a corner — contradicting Proposition 36. Hence E_1 is an a-edge, and the same argument at B gives E_n an a-edge. This is what *(i)* used. ☐

Remark 44. Theorem 43 reduces the whole $e = 1$, $m = 1$ subfamily — which contains the base-β prime candidates $N = 3f^2 - 1$, among them 11, 47, 107, 191, 431, 587 and 971 below 1000 — to the single question of whether that one base walk is realizable. It also corrects a plausible-looking plan: one might hope to close the subfamily by proving $R \geq 2$ on every side. That is false for the base. The walk with $R_{\text{base}} = 2$ is exactly the one the corner parallelogram kills, and the survivor has $R_{\text{base}} = 1$; on the base, "$R \geq 2$" is not a lemma reducing the problem but a restatement of it. On the two equal sides $R \geq 2$ does hold, and is proved above.

Remark 45 (The $e \geq 2$ residue)**.** The $e \geq 2$ residue ($N = 23, 59, 71, 83, 131, 167, \ldots$; the members 23 and 71 are now settled by search, Table 1) resists all three natural invariant methods, and in two cases *provably*:

- **Linear/colouring invariants** — dead by Remark 32.
- **b-edge census.** On a maximal segment of length L the two sides carry $q_1 \equiv q_2 \pmod{f}$ (both $\equiv -e^{-2}L$), and the entire census — in *every* modulus, and in its per-direction refinement — is a formal consequence of the single parity identity $N(a+b+c) \equiv 2X + Y \pmod{2}$, valid for all (e, f, m). So no congruence of this type can bite.
- **Euler / vertex counting.** For any such tiling $V - E + F = 2$ is *identically* the sum of the three corner identities $\#\alpha = \#\beta = \#\gamma = N$; the vertex-type system collapses to $Q_{013} = 1 + P_{320} + 2Q_{640} + Q_{431} + W_{320}$ and $3P_{320} + P_{111} + 6Q_{640} + 4Q_{431} + 2Q_{222} + 3W_{320} + W_{111} = N - 3$, which is feasible for every $e \geq 2$, $m = 1$ target. So Euler carries no information here. Here W_{320} and W_{111} count the interior vertices at which some tile is met in the *relative interior* of one of its edges: such a tile contributes a straight angle, so the remaining tile angles sum to π rather than 2π and the figure is $(3, 2, 0)$ or $(1, 1, 1)$ (Proposition 35). These types are not optional — the dissections here are not edge-to-edge (Remark 21) — and an earlier version of this census omitted them. Their presence only enlarges the feasible set, so the conclusion is unaffected.

What *is* gained is citation-free structure, valid for all (e, f, m): the γ-trap (Proposition 37) and the corner parallelogram (Proposition 40), both proved above from the corner figures and the unsplittability of b, and both used by the exact search as prunes. They were checked against the two genuine certificates before being proved: the first-two-edges rule of Proposition 40 holds on all six sides of the 44- and 99-tilings, and at each of the four base corners exactly one of the two sides begins with a c-edge.

This isolates the residue to one clean statement. Since $q_{\text{base}} \equiv em \pmod f$ while $q_{\text{equal}} \equiv 0$, we have:

if the base carries no b-edge, then $f \mid em$, hence $f \mid m$, hence no tiling with $m = 1$ — for every e at once.

The 44-tiling ($m = 2$) indeed carries *zero* b-edges on its entire boundary, consistent with this. But Remark 33 now shows the converse direction is a dead end: the 99-tiling has $f \nmid m$ and its base carries seven b-edges, so "the base never carries a b-edge" is *false* in general, and the displayed implication — whose hypothesis is equivalent to $f \mid m$ — cannot be used to exclude $m = 1$ without already knowing $f \mid m$, which is false. A fourth method also fails, and provably: *boundary counting*. Each side S of ABC, cut into k_S whole edges, forces at least $2k_S - 1$ tiles to meet it (the k_S inner tiles, plus a distinct "middle" tile at each of the $k_S - 1$ junctions, since a triangle has no two collinear edges); this is exactly tight on the 44-tiling (base $k = 8$, $15 = 2 \cdot 8 - 1$ tiles). But at $m = 1$ the boundary is *minimal* — for $e = 1$ the least feasible total is $\sum k_S = 3f + 2$ — so counting sees only $O(f)$ of the $N = \Theta(f^2)$ tiles, and $\sum(2k_S - 1) = 6f + 1$ exceeds $N = 3f^2 - 1$ only at $f = 2$, the case already closed. The ratio $(6f + 1)/(3f^2 - 1) \to 0$: $m = 1$ is the case *least* constrained by counting, not the most.

So no *invariant* route survives: linear invariants, the census, Euler and counting are all provably vacuous, and the one arithmetic statement that would settle it ($f \mid m$) is now known to be false. What does survive is *structure*. Theorem 12 classifies the boundary walks for every (e, f) at $m = 1$, and in the thin regime $f > 2e$ reduces the whole branch to two possible bases and b-free equal sides — the $e \geq 2$ analogue of Remark 11, and the first statement in this branch that is uniform in e. Parts (iii)–(iv) need no thinness at all: at $m = 1$ *no equal side carries a b-edge*, for every (e, f), so all boundary b-edges lie on the base; and the base's b-count $Q = e + fj$ obeys $j(f - e) \leq e - 1$. It does not close the branch, but it does locate the difficulty exactly. The bound $j \leq (e - 1)/(f - e)$ is uniform, and it degrades precisely as f falls towards e: at $e = 1$ it pins $j = 0$ for every f (one b-edge on the base, the $e = 1$ rigidity of Remark 11, now a corollary); at $f > 2e$ it still pins $j = 0$; but in the *thick* regime $f \leq 2e$ it leaves $\lfloor (e - 1)/(f - e) \rfloor + 1$ values of j and the trichotomy dissolves. That is precisely where the open members sit — $N = 59$ is $(e, f) = (4, 5)$, so $j \leq 3$; $N = 66$ is $(3, 5)$, so $j \leq 1$; likewise $N = 23, 83, 131, 167$. We record the branch as open and settle its members one at a time by exhaustive search. Table 1 lists every member below 110; of the eleven, eight are settled and 59 and 66 are now excluded by exhaustive search (apex-down with the walk prune; 1 838 175 and 7 232 464 nodes), leaving 83 under search. ($N = 83$, $(e, f) = (5, 6)$, is the smallest base-β prime not yet settled.) The walk classification also feeds back into the search: the γ-trap and the corner parallelogram have the multiset corollaries $R \geq 1$ on every side, $Q_{\text{base}} \leq k - 4$ and $Q_{\text{side}} \leq k - 2$ (k the number of edges of the walk), and the engine now prunes any partial boundary walk that fits no surviving multiset (prune P5). The rule was validated positionally on all six sides of the two genuine tilings before use — the 44-tiling's base *aaaaccca* and the 99-tiling's base *aabbbbbbbcc*, which is exactly tight for $Q \leq k - 4$ — the strengthened engine re-finds the 44-tiling from scratch (852,093 nodes; the found tiling passes the independent verifier: congruence, exact area, containment, pairwise disjointness) — and the corner rule is applied only after checking that the tile's b-edge is unsplittable ($n_a a + n_c c = b$ has no solution), which is what forces the parallelogram. For $N = 59$ this cuts the admissible bases from seven to four, for $N = 66$ from four to three.

Table 1. Base-β members with $N \le 110$: every one is settled by exhaustive search (corner-anchored exact search over $\mathbb{Q}(\sqrt{D})$, C++ engine, node-for-node validated against the independent Python engine). "thin" means $f > 2e$, the regime of Theorem 12.

N	(e, f)	tile	regime	search verdict	(nodes)
11	$(1, 2)$	$(2, 3, 4)$	thick	no tiling	135
23	$(2, 3)$	$(6, 5, 9)$	thick	no tiling	19 677
26	$(1, 3)$	$(3, 8, 9)$	thin	no tiling	2 025
39	$(3, 4)$	$(12, 7, 16)$	thick	no tiling	825 724
47	$(1, 4)$	$(4, 15, 16)$	thin	no tiling	12 440
59	$(4, 5)$	$(20, 9, 25)$	thick	no tiling	1 838 175
66	$(3, 5)$	$(15, 16, 25)$	thick	no tiling	7 232 464
71	$(2, 5)$	$(10, 21, 25)$	thin	no tiling	553 417
74	$(1, 5)$	$(5, 24, 25)$	thin	no tiling	132 519
83	$(5, 6)$	$(30, 11, 36)$	thick		searching
107	$(1, 6)$	$(6, 35, 36)$	thin	no tiling	1 251 382

Remark 46 (The colouring theorem is citation-free)**.** The same identity has a positive consequence: it *proves* the sound half of [2, Thm. 14] rather than citing it. Because α/π is irrational (Remark 32's η, via Niven — `tile_alpha_irrational`), η is not a root of unity: $n(\frac{\alpha}{2} + \frac{\pi}{2}) = k\pi$ forces $n = 0$ (`direction_free`, `colouring_well_defined`, both axiom-clean). Hence $\langle \eta, -1 \rangle \cong \mathbb{Z} \times \mathbb{Z}/2$ and the sign character $\chi_2(\pm\eta^n) = \pm(-1)^n$ is *well defined* — which is exactly what makes the colouring number $M = \sum_t \chi_2(d_t)$ a well-defined integer, and the $z = -1$ specialization of the master identity then returns $M = m(f + e)$ together with the size formulas $X = f^3 M/(f + e)$, $Y = eM(3f^2 - e^2)/(f + e)$. Consequently the $3\alpha + 2\beta$ tiling equations — and with them the necessary sides of the triquadratic and four-component branches (Remark 7) — no longer rest on a preprint's colouring theorem: the direction-group foundation is machine-checked, and the cancellation step is the same Φ-style argument proved in Section 4.

Proposition 47 (Base-(α+β) prime exclusion, correcting [2, Thm. 18])**.** *No isosceles triangle with base angles $\alpha + \beta$, where $3\alpha + 2\beta = \pi$ and $\alpha \notin \mathbb{Q}\pi$, is N-tiled for N prime.*

Proof. Fix a putative tiling with coloring number M. On this target [2, Thm. 17] gives $N/M^2 = (f + e)/(f - e) > 1$, hence $0 < M^2 < N$; we use that rather than [2, Cor. 3], which is quoted in the literature without scope and cannot hold in the stated generality (Remark 45). Put $d = \gcd(N - M^2, N + M^2)$. As $d \mid 2N$, $d \mid 2M^2$, and $\gcd(N, M^2) = 1$ (a prime $N > M^2$ cannot divide M^2), $d \mid 2$. Writing $g = (N + M^2)/d$ and $\hat{a} = (N - M^2)/d$, the primitive tile is $(a, b, c) = (g\hat{a},\ g^2 - \hat{a}^2,\ g^2)$ with $0 < \hat{a} < g$ ([2, Thm. 17]; we do not use [2, Lem. 8], whose squarefree half this paper shows to be false — only the parametrization is needed, and it also follows from the $3\alpha + 2\beta$ relation directly), and the base AC has integer length $Y = M\hat{a}(g + \hat{a})$ ([2, Lem. 42]). A tiling forces the base to carry at least *one* c-edge — this is the γ-trap of Remark 45, proved here from the vertex figures alone and machine-checked (`gamma_injection`, `c_edge_exists`) — and a number of b-edges $\equiv -M \pmod g$, hence at least $g - M > 0$ ([2, Lem. 45(iii)]); so $Y \ge (g - M)b + c$. But $d \mid 2$ gives $\hat{a} + M \le g$, and the identity

$$(g - M)(g^2 - \hat{a}^2) - M\hat{a}(g + \hat{a}) = g\,(g + \hat{a})\,(g - \hat{a} - M) \ \ge\ 0$$

gives $Y \le (g - M)b < (g - M)b + 2c$, a contradiction. (E.g. $N = 79,\ M = 1$: $d = 2$, $g = 40$, $\hat{a} = 39$, $b = 79$, $c = 1600$, so $Y = 3081$ while $(g - M)b + 2c = 6281$.) The two arithmetic facts used — $d \mid 2$ and the identity together with $\hat{a} + M \le g$ — are machine-checked in Lean (`lean/BaseAlphaBetaPrime.lean`, axiom-clean); only the covering bounds are cited. □

The tile. Throughout, T has angles (α, β, γ) opposite sides (a, b, c), with $\gamma = 2\pi/3$, hence $\alpha + \beta = \pi/3$; we treat the incommensurable case, α an irrational multiple of π. Once the target is known isosceles (Section 3), T has commensurable sides by [3, Thm. 12.5] (refereed); we scale to

$$(a, b, c) \in \mathbb{Z}^3, \qquad \gcd(a, b, c) = 1, \qquad c^2 = a^2 + ab + b^2 \tag{2}$$

(the law of cosines at $\gamma = 2\pi/3$). We call such a triple a *primitive* $120°$*-triple*. If a prime p divides two of a, b, c it divides the third (by (2)), so (2) gives $\gcd(a, b) = \gcd(a, c) = \gcd(b, c) = 1$. One computes

$$\sin\alpha = \tfrac{a\sqrt{3}}{2c}, \quad \cos\alpha = \tfrac{2b+a}{2c}, \qquad \sin\beta = \tfrac{b\sqrt{3}}{2c}, \quad \cos\beta = \tfrac{2a+b}{2c}. \tag{3}$$

The direction grid. Set $G = \{\, j\frac{\pi}{3} + k\alpha : j, k \in \mathbb{Z}\,\} \pmod{2\pi}$. We claim every edge of every tile in a tiling points in a direction lying in G, for a single fixed reference. Fix the reference along an edge of one tile; its three edge-directions are then in G, since the tile angles $\alpha, \beta = \frac{\pi}{3} - \alpha, \gamma = \frac{2\pi}{3}$ all lie in $\mathbb{Z}\alpha + \mathbb{Z}\frac{\pi}{3}$. Now argue by adjacency. The tiling fills a connected region, so any tile t' shares at least a boundary point P with the union of the tiles already placed; at P the surrounding corner angles each belong to $\{\alpha, \beta, \gamma\}$, and a directed edge of t' at P differs from a known grid direction at P by a sum of such angles, hence lies in G; the remaining edges of t' then differ from it by $\mathbb{Z}$-combinations of $\alpha, \frac{\pi}{3}$, so all of t' is in G. This uses only the angles meeting at a point and so applies verbatim at non-edge-to-edge incidences, where P lies in the relative interior of an edge (the angles there sum to π rather than 2π). Since α/π is irrational, $\frac{\pi}{3}$ and α are $\mathbb{Q}$-linearly independent, so the pair (j, k) is determined by the direction (and $(-1)^j$ by the direction alone, as $2\pi = 6 \cdot \frac{\pi}{3}$).

Vertex types and shapes of ABC. A corner of ABC is filled by tile-corners: if $P, Q, R \ge 0$ (not all 0) of them are α-, β-, γ-corners, the corner measures $P\alpha + Q\beta + R\gamma = (P - Q)\alpha + (Q + 2R)\frac{\pi}{3}$. Write $(m, k) := (P - Q,\ Q + 2R)$. Since $Q \equiv k \pmod 2$, $0 \le Q \le k$, and $P = m + Q \ge 0$, the pair (m, k) is *realizable* exactly when

$$k \ge 1 \text{ and } m \ge -k, \qquad \text{or} \qquad k = 0 \text{ and } m \ge 1. \tag{4}$$

The three corners of ABC have angles $m_i\alpha + k_i\frac{\pi}{3}$ summing to $\pi = 3 \cdot \frac{\pi}{3}$; as α is irrational this forces

$$\textstyle\sum_i m_i = 0, \qquad \sum_i k_i = 3. \tag{5}$$

By (4) each $m_i + k_i \ge 0$, and by (5) $\sum_i (m_i + k_i) = 3$; hence each $m_i + k_i \in \{0, 1, 2, 3\}$ and $m_i \in [-k_i,\ 3 - k_i] \subseteq [-3, 3]$. The enumeration is therefore finite and *independent of* α. Checking the finitely many triples $\{(m_i, k_i)\}$ satisfying (4), (5) and $0 < m_i\alpha + k_i\frac{\pi}{3} < \pi$ (done in exact arithmetic in Section 8) leaves exactly eleven shapes of ABC: the equilateral triangle; two isosceles triangles, with base angles α (apex $\alpha + 3\beta$) and β (apex $\beta + 3\alpha$); and eight scalene triangles, namely

$$F_1 = (\alpha, \alpha + \beta, \alpha + 2\beta), \quad F_2 = (2\alpha, 2\beta, \alpha + \beta), \quad F_3 = (\alpha, 2\alpha, 3\beta), \quad F_4 = (\alpha, 2\beta, 2\alpha + \beta),$$

their reflections under $\alpha \leftrightarrow \beta$ (with F_2 self-paired), and the tile-similar triangle $(\alpha, \beta, 2\pi/3)$. Since α/π is irrational, distinct pairs (m, k) give distinct angles, so the labels *equilateral*, *isosceles*, *scalene* attached to these types are unambiguous: no accidental coincidence of angles can occur, as any such coincidence would force $\alpha \in \mathbb{Q}\pi$. These types coincide with the six sporadic non-equilateral $2\pi/3$ similarity types of Zhang [7].

The area identity. If ABC is similar to a fixed shape whose primitive integer side-proportion vector is D (gcd of its entries 1), and ABC is N-tiled, then each side of ABC is a disjoint union of whole tile-edges (a tile meeting a straight portion of the boundary has a whole side on it), hence an integer; so the sides of ABC are $k \cdot D$ for an integer $k \geq 1$, and

$$N = \frac{\mathrm{Area}(ABC)}{\mathrm{Area}(T)} = k^2 N_0, \qquad N_0 := \frac{\mathrm{Area}(D)}{\mathrm{Area}(T)}, \qquad \mathrm{Area}(T) = \tfrac{\sqrt{3}}{4}ab. \tag{6}$$

This is a necessary condition on N.

3. Reduction to the isosceles case

The reduction below takes the tile to be integer-sided. That input is a citation ([3, Thm. 12.5] on the isosceles targets, [6] in general). The following proposition recovers it from the invariant alone, with no citation, on any target where the invariant takes integer values — which is how it is used in Sections 4 onwards. We state it first because it is logically prior to everything that follows.

Proposition 48 (Rationality from integrality of the invariant)**.** *Let T have sides a, b, c with $c^2 = a^2 + ab + b^2$ (equivalently, the angle opposite c is $2\pi/3$), let $P \neq 0$ be a real scale, and suppose the three quantities*

$$N = P^2\, b(a + 2b), \qquad M_\alpha = P(c - a - b), \qquad M_\beta = P(c + a + b)$$

are all integers. If $b \neq 0$ and $a + b \neq 0$, then $a/b \in \mathbb{Q}$ and $c/b \in \mathbb{Q}$: the tile is rational up to scale.

Proof. Two identities. First, $M_\alpha M_\beta = P^2\big(c^2 - (a + b)^2\big) = -P^2 ab$ by the tile relation, whence

$$N + M_\alpha M_\beta = P^2 b(a + 2b) - P^2 ab = 2P^2 b^2 \neq 0,$$

and $a\,(N + M_\alpha M_\beta) = 2P^2 ab^2 = b\,(-2M_\alpha M_\beta)$, so

$$\frac{a}{b} = \frac{-2M_\alpha M_\beta}{N + M_\alpha M_\beta} \in \mathbb{Q}.$$

Second, $M_\beta - M_\alpha = 2P(a + b) \neq 0$ and $M_\alpha + M_\beta = 2Pc$, so $c\,(M_\beta - M_\alpha) = (a + b)(M_\alpha + M_\beta)$ and

$$\frac{c}{a + b} = \frac{M_\alpha + M_\beta}{M_\beta - M_\alpha} \in \mathbb{Q}.$$

Multiplying by $(a+b)/b = a/b+1 \in \mathbb{Q}$ gives $c/b \in \mathbb{Q}$. Machine-checked in `lean/Rationality.lean` (`tile_rational`), axiom-clean. ∎

Remark 49. The proposition converts the rationality input from a hypothesis about the tile into a hypothesis about the invariant, and the latter is what the method produces natively: wherever the boundary functional of Section 4 is defined and integral, rationality is automatic. It does not by itself discharge the citation on the *general* target, because establishing integrality of all three quantities there is exactly the content of the cited theorem. What it does discharge is the appearance of a rationality assumption *inside* the invariant argument, where the three integralities are already in hand; and it shows the two facts are equivalent rather than independent, which is not how the literature presents them.

Proposition 50. *Let T be a tile with angles $(\alpha, \beta, 2\pi/3)$, α/π irrational. If T N-tiles a triangle and N is prime, then ABC is isosceles and not equilateral.*

Proof. By Section 2, the corner-type triple of ABC is the equilateral one, the tile-similar one, one of $F_1, \dots, F_4$ or their reflections, or one of the two isosceles ones. The equilateral case gives N not prime [5, Thm. 6]; the tile-similar case gives $N = n^2$ [1, 12], not prime. For the remaining non-isosceles shapes we compute N_0 from (6); in each case $N = k^2 N_0$ is composite.

F_2 and F_4. The side-proportion vector of F_2 is $\big(a(a+2b),\, b(2a+b),\, c^2\big)$ with the $\pi/3$ vertex between the first two sides, giving $N_0 = (a+2b)(2a+b)$. For F_4 one checks the side-vector $\big(ac,\, b(2a+b),\, (a+b)c\big)$ has gcd 1 (any common prime g divides c and then $g^2 \mid 3b^2$, forcing $g = 1$), giving $N_0 = (a+b)(2a+b)$. Both are products of two integer factors each at least 3, hence composite, so $N = k^2 N_0$ is composite.

F_3. The law of sines gives the side-proportion vector $\big(ac^2,\, a(a+2b)c,\, 3ab(a+b)\big)$, whose entries share the factor a; dividing it out leaves $\big(c^2,\, (a+2b)c,\, 3b(a+b)\big)$, and $N_0 = 3(a+b)(a+2b)$. This is divisible by 3, hence composite, so $N \equiv 0 \pmod 3$ is composite.

F_1. The side-vector is $(a, c, a+b)$ with the $\pi/3$ vertex between the sides a and $a+b$, so $N_0 = (a+b)/b$ and $N = k^2(a+b)/b$. As $\gcd(b, a+b) = 1$, integrality forces $b \mid k^2$, so $N = t(a+b)$ with $t = k^2/b \in \mathbb{Z}_{\geq 1}$. It remains to show $a+b$ is composite. Suppose $a+b = p$ is prime. Since $(a+b)^2 - c^2 = ab$ by (2), we have $ab = p^2 - c^2$, so a, b are the roots of $x^2 - px + (p^2 - c^2)$, whose discriminant $4c^2 - 3p^2$ must be a perfect square d^2; then $3p^2 = (2c-d)(2c+d)$. With $2c - d < 2c + d$ the factorizations of $3p^2$ into two positive factors are $(1, 3p^2)$, $(3, p^2)$, $(p, 3p)$, and in each the smaller root $\frac{p-d}{2}$ is not a positive integer: in the first, $d = (3p^2-1)/2 > p$, so the root is negative; in the second, $d = (p^2-3)/2$ and the root $\frac{p-d}{2} = -\frac{(p-3)(p+1)}{4}$ is negative for $p > 3$; in the third, $d = p$ gives $c = p$ and $ab = p^2 - c^2 = 0$. (The case $p \leq 3$ is impossible since $a + b \geq 8$ for a primitive 120°-triple.) Hence $a+b \geq 8$ is composite, and $N = t(a+b)$ is composite.

The reflected shapes are handled by $\alpha \leftrightarrow \beta$ (which swaps $a \leftrightarrow b$ and leaves each N_0 above of the same form). So a prime N forces the corner-type triple of ABC to be one of the two isosceles types; those have angles $(\varphi, \varphi, \pi - 2\varphi)$ with $\varphi \in \{\alpha, \beta\}$, and they are never equilateral ($\varphi = \frac{\pi}{3}$ would force $\beta = 0$ or $\alpha = 0$). Hence ABC is isosceles and not equilateral. □

4. The signed-direction invariant

Fix the reference so the direction grid is $G = \{\, j\frac{\pi}{3} + k\alpha \,\}$. Because $\beta = \frac{\pi}{3} - \alpha$, every grid direction also has the form $(j+k)\frac{\pi}{3} - k\beta$, so we may read off the coefficient of $\frac{\pi}{3}$ in two coordinate systems. Define, for a direction $\theta \in G$,

$$f_\alpha(\theta) = (-1)^j \quad (\theta = j\tfrac{\pi}{3} + k\alpha), \qquad f_\beta(\theta) = (-1)^{j'} \quad (\theta = j'\tfrac{\pi}{3} + k'\beta).$$

Both are well defined (Section 2), and both satisfy

$$f(\theta + \pi) = -f(\theta), \tag{7}$$

since adding $\pi = 3 \cdot \frac{\pi}{3}$ increases the $\frac{\pi}{3}$-coefficient by 3. They are genuinely different functions: $f_\beta = (-1)^k f_\alpha$.

For $f \in \{f_\alpha, f_\beta\}$ and a directed edge of length L and direction θ, assign the weight $L\, f(\theta)$. For a tile t with boundary traversed counterclockwise, put $C_f(t) = \sum_{\text{edges}} L\, f(\theta)$, and for the boundary of ABC put $\Phi_f(\partial ABC) = \sum_{\text{sides}} L\, f(\theta)$ (counterclockwise).

Lemma 51 (Cancellation)**.** *For each $f \in \{f_\alpha, f_\beta\}$, $\displaystyle\sum_{\text{tiles}} C_f(t) = \Phi_f(\partial ABC)$.*

Proof. For a grid direction θ let $\Lambda(\theta)$ be the total length of all directed tile-edges of direction θ, summed over every tile (counterclockwise) and every edge; then, grouping the double sum

$\sum_t C_f(t)$ by direction,

$$\sum_{\text{tiles}} C_f(t) = \sum_{\theta} \Lambda(\theta)\, f(\theta). \tag{8}$$

Split each tile-edge into its maximal sub-segments that are either interior to ABC or contained in ∂ABC; this only refines the sums and changes nothing. Consider the finite set of maximal straight segments S carved out on the tiling's edges. If S lies in the interior of ABC, a neighbourhood of a generic point of S is a disc split by S into two half-discs, each tiled; the tiles on one side meet S in directed edges of total length $|S|$ pointing one way, say θ, and those on the other side in directed edges of total length $|S|$ pointing $\theta+\pi$ (the two subdivisions may differ, a non-edge-to-edge incidence, but each side covers S exactly once because the tiles cover ABC with disjoint interiors). If instead $S \subseteq \partial ABC$, only the single tile inside ABC meets it, contributing length $|S|$ in the counterclockwise direction of ∂ABC. Writing $\Lambda = \Lambda_{\text{int}} + \Lambda_{\text{bd}}$ accordingly, interior segments give $\Lambda_{\text{int}}(\theta) = \Lambda_{\text{int}}(\theta+\pi)$ for every θ, while $\Lambda_{\text{bd}}(\theta)$ is the counterclockwise boundary length in direction θ. Substituting into (8) and pairing θ with $\theta+\pi$,

$$\sum_{\{\theta,\theta+\pi\}} \Lambda_{\text{int}}(\theta)\big(f(\theta)+f(\theta+\pi)\big) = 0$$

by (7), so $\sum_{\text{tiles}} C_f(t) = \sum_\theta \Lambda_{\text{bd}}(\theta) f(\theta) = \Phi_f(\partial ABC)$. $\square$

Lemma 52 (Tile value)**.** *For every placement of T, $C_{f_\alpha}(t) = \pm(c+a-b)$ and $C_{f_\beta}(t) = \pm(c+b-a)$.*

Proof. Take a positively oriented placement and label the vertices V_1, V_2, V_3 in counterclockwise order with interior angles α, β, γ respectively; let e_i be the directed edge from V_i to V_{i+1} (indices mod 3). Then e_1, e_2, e_3 are opposite V_3, V_1, V_2, so they have lengths c, a, b. Traversing counterclockwise turns left at V_{i+1} by the exterior angle $\pi - (\text{angle at } V_{i+1})$, so $\operatorname{dir}(e_2) = \operatorname{dir}(e_1) + (\pi-\beta)$ and $\operatorname{dir}(e_3) = \operatorname{dir}(e_2) + (\pi - \gamma)$. Now π adds 3 to the $\frac{\pi}{3}$-coefficient j, while α adds 0, $\beta = \frac{\pi}{3} - \alpha$ adds 1, and $\gamma = \frac{2\pi}{3}$ adds 2; so $\pi - \beta$ adds $3-1=2$ and $\pi-\gamma$ adds $3-2=1$. Writing j_0 for the $\frac{\pi}{3}$-coefficient of $\operatorname{dir}(e_1)$, the three coefficients are j_0, j_0+2, j_0+3, so $f_\alpha = (-1)^{j_0}$ on e_1, e_2 and $-(-1)^{j_0}$ on e_3. As e_1, e_2, e_3 carry lengths c, a, b,

$$C_{f_\alpha}(t) = (-1)^{j_0}\big(c+a-b\big).$$

Any placement is obtained from this one by a grid rotation $m\alpha + n\frac{\pi}{3}$, which shifts every j by n and multiplies C_{f_α} by $(-1)^n$, possibly followed by a reflection, which reverses the traversal and negates C_{f_α}; hence $C_{f_\alpha}(t) = \pm(c+a-b)$ in every case. Repeating in the β-frame (where $\gamma = \frac{2\pi}{3}$ adds 0 to j' and α adds 1) makes the a-edge the odd one out, giving $C_{f_\beta}(t) = \pm(c+b-a)$. $\square$

Corollary 53 (Integrality and parity)**.** *Each of $M_\alpha = \dfrac{\Phi_{f_\alpha}(\partial ABC)}{c+a-b}$ and $M_\beta = \dfrac{\Phi_{f_\beta}(\partial ABC)}{c+b-a}$ is an integer, and moreover*

$$M_\alpha \equiv M_\beta \equiv N \pmod 2.$$

They are separate necessary conditions: each follows from Lemma 51 for its own f, so a single failure forbids a tiling.

Proof. By Lemmas 51 and 52, $\Phi_{f_\alpha}(\partial ABC) = \sum_t C_{f_\alpha}(t) = (c+a-b)\sum_t \varepsilon_t$ with $\varepsilon_t \in \{\pm 1\}$, so $M_\alpha = \sum_t \varepsilon_t$ is a sum of N signs, whence $M_\alpha \equiv N \pmod 2$; likewise for f_β. $\square$

5. The isosceles obstruction

Lemma 54 (Non-integrality)**.** *Let a, k be positive integers with $\gcd(a, k) = 1$, put $b = k^2$, and let $c > 0$ satisfy $c^2 = a^2 + ab + b^2$. Then $k \nmid (a + b - c)$.*

Proof. From $c^2 = a^2 + ab + b^2$ we get $c > a$ and, since $c^2 < (a + b)^2$, also $c < a + b$. Suppose $k \mid (a + b - c)$. As $b = k^2 \equiv 0 \pmod k$, this gives $c \equiv a \pmod k$; with $0 < c - a < b = k^2$ we may write $c = a + kt$, $1 \le t \le k - 1$. Substituting into $c^2 = a^2 + ak^2 + k^4$ and dividing by k,

$$2at + kt^2 = ak + k^3, \qquad \text{that is} \qquad a(2t - k) = k(k - t)(k + t).$$

Then $k \mid a(2t - k)$, and $\gcd(a, k) = 1$ gives $k \mid (2t - k)$, hence $k \mid 2t$. But $2 \le 2t \le 2k - 2$ contains no multiple of k other than k itself, so $2t = k$. The left side is then $a(2t-k) = 0$, while the right side is $k \cdot \frac{k}{2} \cdot \frac{3k}{2} = \frac{3k^3}{4} > 0$, a contradiction. Hence $k \nmid (a + b - c)$. □

Theorem 55. *Let T be a tile with angles $(\alpha, \beta, 2\pi/3)$, α/π irrational. Then no prime number of copies of T tiles an isosceles, non-equilateral triangle.*

Remark 56. The irrationality hypothesis cannot be dropped: the commensurable tile $(\frac{\pi}{6}, \frac{\pi}{6}, \frac{2\pi}{3})$ tiles an equilateral (hence isosceles) triangle with $N = 3$, by the fan from the centre; that tile lives in the commensurable branch of the classification, where $N = 3n^2$ is allowed. The equilateral target for an *incommensurable* $2\pi/3$ tile is excluded for primes by Beeson [5], and is covered in the assembly of Section 6; it is not part of this theorem.

Proof. ABC has exactly two equal corner angles. Each corner angle is $m\alpha + k\frac{\pi}{3}$ (Section 2); since α/π is irrational, two corner angles are equal only if their pairs (m, k) coincide. So the corner-type triple of ABC has exactly one repeated type, and among the eleven triples of Section 2 exactly the two isosceles ones qualify. Hence the base angle is α or β. We use the invariant generated by the base angle, so that the equal sides land on $\frac{\pi}{3}$-coefficient 3 (odd) and pick up $f = -1$.

If the base angle is α, then $ABC = k(c, c, 2b + a)$ (equal sides kc, base $k(2b + a)$): the base is at direction 0 ($f_\alpha = +1$) and the equal sides at $\pi - \alpha = 3\frac{\pi}{3} - \alpha$ ($f_\alpha = -1$), so $\Phi_{f_\alpha}(\partial ABC) = k(2b+a) - 2kc = k(2b+a-2c)$. The tiling count is $N = k^2(a+2b)/b$; since N is prime and $\gcd(a+2b, b) = 1$, we have $k^2 = b$, i.e. $k = \sqrt{b}$. By Corollary 53 and $c^2 = a^2+ab+b^2$,

$$\frac{k(2b + a - 2c)}{c + a - b} = \frac{c - a - b}{k} \in \mathbb{Z}, \qquad k = \sqrt{b},$$

so $k \mid (a + b - c)$, contradicting Lemma 54. If the base angle is β, then $ABC = k(c, c, 2a + b)$, the equal sides are at $\pi - \beta$, and using f_β (for which the equal sides have $\frac{\pi}{3}$-coefficient 3) the same computation gives, with $k = \sqrt{a}$, $(c - a - b)/k \in \mathbb{Z}$, again contradicting Lemma 54 (the integer $a + b - c$ is symmetric in a, b). In both cases no tiling exists. □

Remark 57. The role of the two invariants is essential. For the base-angle-β shape, f_α places the equal sides at $\pi - \beta = \frac{2\pi}{3} + \alpha$, an *even* $\frac{\pi}{3}$-coefficient, so f_α gives an integer value and does not obstruct; f_β does. Each invariant is a necessary condition in its own right (Corollary 53), so using the one that fails is legitimate, not a choice of convenient frame: on any genuine tiling both invariants return an integer.

6. Resolution of the problem

Proof of Theorem 1. Let $p > 3$ be prime with $p \equiv 3 \pmod 4$ and $p \ne 3f^2 - e^2$ for every coprime $1 \le e < f$, and suppose some triangle is p-tiled. By the classification (Section 2) the tile is in one of the listed types. Consider the types other than a $2\pi/3$ tile on a non-equilateral triangle. For the $3\alpha + 2\beta$ branch, four of its five targets are excluded outright — the two scalene ones by the machine-checked cores of [2, Thms. 8, 12], the base-$(\alpha+\beta)$ target by Proposition 47 and the base-α target by Proposition 8 (the last two replacing [2, Thms. 18, 20], whose printed proofs are unsound). Its fifth target, base-β, is *not* excluded by any sound general argument (Remark 9); but its candidate counts are exactly the $3f^2 - e^2$ with $\gcd(e, f) = 1$, which the hypothesis excludes. The remaining branches exclude such a p by existing theorems [12, 1, 4, 3, 5]. For a $2\pi/3$ tile, Proposition 50 forces ABC isosceles and not equilateral, and Theorem 55 excludes that. Hence no such tiling exists. Finally $19 > 3$ is prime, $\equiv 3 \pmod 4$, and is not a base-β candidate: for $f \le 2$ one has $3f^2 - e^2 \le 11$; for $f = 3$, $3f^2 - e^2 \in \{26, 23, 18\}$; and for $f \ge 4$, $3f^2 - e^2 \ge 2f^2 + 2f - 1 \ge 39$. So no triangle is cut into 19 congruent triangles. □

7. Towards the full problem

Erdős's question asks for *all* N. Theorem 1 settles the negative side for primes; this section records what the present methods contribute to the full determination: a complete answer for primes, and, for each sporadic $2\pi/3$ branch, a determination of its *admissible spectrum* — the set of counts compatible with the area and invariant conditions.

Theorem 58 (Fibonacci families are the extremal ones)**.** *For a base-β family (e, f) with $\gcd(e, f) = 1$, $1 \le e < f$, write $M = f + e$ and $N_0 = 3f^2 - e^2$. Then*

$$M^2 - N_0 = -2\,(f^2 - ef - e^2),$$

so $|M^2 - N_0| \ge 2$, with equality if and only if $f^2 - ef - e^2 = \pm 1$, that is, if and only if (e, f) are consecutive Fibonacci numbers. In that case, writing $(e, f) = (F_n, F_{n+1})$,

$$M = F_{n+2}, \qquad N_0 = F_{n+2}^2 - 2(-1)^{n+1},$$

so the counts are exactly a Fibonacci square displaced by 2:

$$11 = 3^2{+}2, \quad 23 = 5^2{-}2, \quad 66 = 8^2{+}2, \quad 167 = 13^2{-}2, \quad 443 = 21^2{+}2, \quad 1154 = 34^2{-}2, \ \dots$$

Proof. $(f + e)^2 - (3f^2 - e^2) = -2f^2 + 2ef + 2e^2 = -2(f^2 - ef - e^2)$. For coprime $e < f$ the integer $f^2 - ef - e^2$ is nonzero, whence $|M^2 - N_0| \ge 2$. Equality means $f^2 - ef - e^2 = \pm 1$, which is the Fibonacci characterisation: the solutions of $x^2 - xy - y^2 = \pm 1$ in positive integers are exactly the consecutive Fibonacci pairs. Then $M = F_n + F_{n+1} = F_{n+2}$ and $f^2 - ef - e^2 = (-1)^n$ give the displayed formula. □

Remark 59 (The field of computation)**.** For the tile $(ef, f^2 - e^2, f^2)$ one has $\sin\alpha = e\sqrt{D}/(2f^2)$ with $D = 4f^2 - e^2 = (2f - e)(2f + e)$, so a base-$\beta$ tiling's coordinates lie in $\mathbb{Q}(\sqrt{D})$. This is the field the exact search computes in; for $(e, f) = (1, 2)$ it is $\mathbb{Q}(\sqrt{15})$, the ring of all three tiling certificates.

Theorem 60 (Scaling ladder)**.** *Let a triangle T be cut into N copies of a tile t. Then for every integer $k \ge 1$ the scaled triangle kT is cut into $k^2 N$ copies of t.*

Proof. Subdivide kT by the triangular grid of side k: it decomposes into k^2 triangles, of which $\binom{k+1}{2}$ are translates of T and $\binom{k}{2}$ are its point-reflections, all congruent to T. Cut each into N copies of t. □

Corollary 61 (The realizable set of a base-β family is a union of multiples)**.** *Fix coprime $1 \le e < f$ and put $N_0 = 3f^2 - e^2$. Since the base-β target at scale m is $m \cdot (f^3, f^3, eN_0)$ — the same shape for every m — the set*

$$S(e, f) = \{\, m \ge 1 : \text{ the scale-}m\text{ target is cut into } m^2 N_0 \text{ copies of } (ef, f^2 - e^2, f^2)\,\}$$

satisfies $m \in S(e, f) \Rightarrow km \in S(e, f)$ for every $k \ge 1$. Hence $S(e, f)$ is the union of the multiples of its minimal elements, and $1 \in S(e, f)$ is not *implied by $S(e, f) \ne \emptyset$.*

Corollary 62 (An infinite family of tilings in an incommensurable branch)**.** *For $(e, f) = (1, 2)$, tile $(2, 3, 4)$ and $N_0 = 11$: $1 \notin S$ (Table 1: the target $(8, 8, 11)$ is exhausted at 135 nodes), while $2 \in S$ and $3 \in S$ by the explicit 44- and 99-tilings. Therefore $S \supseteq 2\mathbb{Z}_{\geq 1} \cup 3\mathbb{Z}_{\geq 1}$, and*

$$N = 11m^2 \text{ is a number of congruent triangles for every } m \text{ divisible by } 2 \text{ or } 3: \quad 44,\ 99,\ 176,\ 396,\ 704,\ 891,\ 110$$

Moreover the 44- and 99-tilings are primitive*: neither arises from Theorem 60 applied to a smaller member of its own family, because that would require $1 \in S$. The values $N = 11m^2$ with $\gcd(m, 6) = 1$ — the smallest being 275 — are undetermined.*

Remark 63. Two points. First, the exclusion of 11 together with the realizability of $11 \cdot 2^2$ and $11 \cdot 3^2$ shows that a base-β family can be empty at scale 1 and non-empty above it; the obstruction at $m = 1$ is genuinely a small-scale phenomenon and not a property of the family. Second, before this the smallest tiling known in any incommensurable branch had 1215 tiles; the corollary supplies infinitely many, of which 44 is the smallest. Whether $S(e, f) \neq \emptyset$ for every (e, f) — equivalently, whether every base-β family tiles at *some* scale — is open; the first untested instances are the scale-2 targets of the families $(2, 3)$, $(3, 4)$ and $(1, 4)$, with $N = 92$, 156 and 188.

Theorem 64 (The prime case)**.** *Let p be a prime with $p \not\equiv 11 \pmod{12}$. Then p is a number of congruent triangles into which some triangle can be cut if and only if*

$$p = 2, \qquad p = 3, \qquad \text{or} \qquad p \equiv 1 \pmod 4.$$

The same equivalence holds for those $p \equiv 11 \pmod{12}$ that have been settled by exhaustive search, currently $11, 23, 47, 59, 71, 107$ (Table 1); for the remaining primes $\equiv 11 \pmod{12}$ — the smallest being 83 — the "only if" direction is open.

Remark 65 (The prime case of Erdős's problem, in one line)**.** Combining Theorem 2 with the branch analysis, the prime part of the problem has been reduced to a single congruence class. A prime p lies in the spectrum if $p = 2$, $p = 3$ or $p \equiv 1 \pmod 4$ (classical constructions), and does not if $p \equiv 7 \pmod{12}$ (Corollary 3); the class $3 \pmod{12}$ contains only $p = 3$. What remains is exactly

$$p \equiv 11 \pmod{12},$$

which is precisely the set of base-β candidates, conjecturally empty of tile counts, and verified so for $11, 23, 47, 59, 71, 107$. By Dirichlet the resolved and unresolved halves of the primes $\equiv 3 \pmod 4$ have equal density, so the prime problem is settled for a set of primes of density $\frac{3}{4}$ among all primes, and open on a set of density $\frac{1}{4}$.

Proof. If. For $p = 2$, the altitude from the right angle splits an isosceles right triangle into two congruent triangles. For $p = 3$, join the centre of an equilateral triangle to its vertices. For $p \equiv 1 \pmod 4$, Fermat's two-squares theorem gives $p = e^2 + f^2$ with $e, f \geq 1$; splitting a right triangle with legs e, f by the altitude from the right angle and tiling the two similar pieces quadratically with e^2 and f^2 congruent copies realizes $N = e^2 + f^2$ (the biquadratic tiling [13, 3]). *Only if.* Theorem 1, whose dependencies this direction inherits. □

Theorem 66 (Isosceles admissible spectrum)**.** *Let (a, b, c) be a primitive $120°$-triple and write $b = de^2$ with d squarefree. Every N-tiling of the base-α isosceles target has scale $k = dew$ for a positive integer w, and*

$$N = d\,w^2\,(a + 2b), \qquad e \mid w\,(c - a - b), \qquad \frac{w(c - a - b)}{e} \equiv N \pmod 2.$$

Symmetrically, with $a = d'e'^2$, the base-β target forces $N = d'w^2(2a + b)$, $e' \mid w(c - a - b)$ and the same parity for $w(c - a - b)/e'$.

Proof. By the area identity $Nb = k^2(a+2b)$ and $\gcd(a+2b, b) = 1$ we get $b \mid k^2$. For a prime $p \mid d$ the valuation $v_p(b) = 2v_p(e) + 1 \le 2v_p(k)$ forces $v_p(k) \ge v_p(e) + 1$, and for $p \nmid d$, $v_p(k) \ge v_p(e)$; hence $de \mid k$, say $k = dew$, and substituting back, $N = dw^2(a + 2b)$. By Corollary 53 and the identity $(c - a - b)(c + a - b) = b(2b + a - 2c)$, the invariant value is $M = k(c - a - b)/b$, so $b \mid k(c - a - b)$, i.e. $de^2 \mid dew(c - a - b)$, i.e. $e \mid w(c - a - b)$; and $M = w(c - a - b)/e \equiv N \pmod 2$ by Corollary 53. □

Theorem 55 is the degenerate case: N prime forces $d = w = 1$, so $b = e^2$ and $e \mid (c - a - b)$, which Lemma 54 refutes. The residual divisibility is governed by the following classification.

Proposition 67 (Solvability of $e \mid (a + b - c)$)**.** *Let $b = de^2$, d squarefree, $\gcd(a, b) = 1$. Then $e \mid (a+b-c)$ holds for a primitive $120°$-triple if and only if there is an integer j with $d < j < 2d$ such that ej is even,*

$$a = \frac{e^2(2d - j)(2d + j)}{4(j - d)}$$

is a positive integer coprime to de^2; in that case $c = a + \frac{e^2 j}{2}$. In particular, for $d = 1$ the range $(1, 2)$ contains no integer, which re-proves Lemma 54; for $d = 2$, $j = 3$, $e = 2$ one obtains $(a, b, c) = (7, 8, 13)$.

Proof. Since $e \mid b$, the hypothesis gives $c \equiv a \pmod e$, and $0 < c - a < b$ writes $c = a + et$ with $1 \le t \le de - 1$. Substituting into $c^2 = a^2 + ab + b^2$ and dividing by e, $a(2t - de) = e(de - t)(de + t)$. As $e \mid a(2t - de)$ and $\gcd(a, e) = 1$, we get $e \mid 2t$, say $2t = ej$ with $1 \le j \le 2d - 1$; substituting $t = ej/2$ gives $4a(j - d) = e^2(2d - j)(2d + j)$, whose right side is positive, forcing $j > d$ and the displayed formula. Conversely, given such j, setting $t = ej/2$, a as displayed and $c = a + et$ reverses the computation: $c^2 = a^2 + ab + b^2$ and $a + b - c = e(de - t)$ is divisible by e. □

Theorem 68 (Spectrum lattice)**.** *Let (a, b, c) be a primitive $120°$-triple, $b = de^2$ with d squarefree, $r = c - a - b$, $g = \gcd(e, r)$, $e_1 = e/g$, and*

$$T \;=\; \frac{r}{g} + d\, e_1^2\, (a + 2b) \pmod 2.$$

Set $E = e_1$ if T is even and $E = 2e_1$ if T is odd. Then the counts admissible for the base-α isosceles target (Theorem 66 with the parity clause) are exactly

$$N = d\,(Eu)^2\,(a + 2b), \qquad u = 1, 2, 3, \ldots$$

Proof. By Theorem 66 the conditions on w are $e \mid wr$ and $wr/e \equiv N \pmod 2$. As $\gcd(e_1, r/g) = 1$, the divisibility is equivalent to $e_1 \mid w$; writing $w = e_1 u$, the count is $M = ur/g$, and using $u^2 \equiv u \pmod 2$,

$$M - N = \frac{ur}{g} - d\, e_1^2 u^2 (a + 2b) \equiv u\Big(\frac{r}{g} + d\, e_1^2 (a + 2b)\Big) = uT \pmod 2,$$

so the parity clause holds for all u when T is even, and exactly for even u when T is odd. □

Theorem 68 uses only the base-aligned invariant. The mirror invariant is also a necessary condition on the same target, and it closes the spectrum completely.

Theorem 69 (Spectrum theorem)**.** *Let (a, b, c) be a primitive $120°$-triple. The tile counts satisfying* all *the invariant conditions on the base-α isosceles target — integrality and parity of both M_α and M_β, together with the area equation — are exactly Zhang's family*

$$N \;=\; m^2\, b\, (a + 2b), \qquad m = 1, 2, 3, \ldots$$

Symmetrically, the base-β target admits exactly $N = m^2 a(2a + b)$, and the F_1 target exactly $N = m^2 b(a + b)$.

Proof. Write $X = c+a-b$, $Y = c+b-a$, so $XY = 3ab$ by (2). On the base-α target with scale k, $M_\alpha = k(2b+a-2c)/X$ and $M_\beta = k(2b+a+2c)/Y$ (the equal sides carry $\frac{\pi}{3}$-coefficient 3 in the α-frame and 2 in the β-frame). Since $(2b+a-2c)+2X = 3a$ and $(2b+a+2c)-2Y = 3a$, the two integralities are equivalent to

$$X \mid 3ak \qquad \text{and} \qquad Y \mid 3ak,$$

and, setting $A = 3ak/X$, $B = 3ak/Y$, the parity clauses of Corollary 53 read $A \equiv B \equiv N \pmod 2$. Recall $k = dew$ and $N = dw^2(a+2b)$, with $b = de^2$.

If $e \mid w$, say $w = em$, then $k = de^2m = bm$, so $A = Ym$ and $B = Xm$ are integers, and $A - N \equiv m\big(Y + b(a+2b)\big) \pmod 2$ using $m^2 \equiv m$; now $Y + b(a+2b) \equiv c+a+b+ab \equiv 0 \pmod 2$, because (2) gives $c \equiv a+b+ab$; likewise for B. So every member of Zhang's family is admissible.

Conversely, fix a prime p and let $x, y, \sigma, \varepsilon, \delta, \omega$ be the p-adic valuations of $X, Y, 3a, e, d, w$; then $x+y = \sigma+\delta+2\varepsilon$ and the divisibilities give $\max(x,y) \le \sigma+\delta+\varepsilon+\omega$, so $\omega \ge \varepsilon - \min(x,y)$. A common prime of X and Y divides $X+Y = 2c$ and $Y-X = 2(b-a)$; for odd p this forces $p \mid c$ and $p \mid b-a$, whence $3 \mid c$ or $p \mid \gcd(a,b)$ — both impossible ($3 \nmid c$ as in Proposition 50). So $\min(x,y) = 0$ for every odd p, and $\omega \ge \varepsilon$ there. At $p = 2$: if ab is odd then X, Y are odd and the same argument applies. If b is even (so a is odd), a two-line computation modulo 8 shows $v_2(b) \notin \{1,2\}$: $b \equiv 2 \pmod 4$ gives $c^2 \equiv 1+2 \equiv 3 \pmod 4$, and $v_2(b) = 2$ gives $c^2 \equiv 1+4 \equiv 5 \pmod 8$, both impossible. So $v_2(b) \ge 3$. Here $v_2(X+Y) = v_2(2c) = 1$ forces $\min(x,y) = 1$, and the divisibilities alone allow $\omega = \varepsilon - 1$; but then exactly one of A, B is odd (their 2-valuations are 0 and $v_2(b)-2$), while $v_2(N) = \delta_2 + 2\omega = v_2(b) - 2 \ge 1$ makes N even — so the parity clause fails, and $\omega \ge \varepsilon$ at $p = 2$ as well. Hence $e \mid w$, i.e. $N = m^2 b(a+2b)$. The base-β case is the mirror $a \leftrightarrow b$. For F_1 the counts are $M_\alpha = 3ak/X - k$ and $M_\beta = 3ak/Y + k$ (Section 7), so the same divisibilities appear, the parity clauses read $A \equiv B \equiv N + k \pmod 2$, and in the boundary case $v_2(k) = \delta_2 + 2\varepsilon_2 - 1 \ge 2$ makes $N+k$ even against an odd count; the argument is otherwise identical, giving $N = m^2 b(a+b)$. ☐

Remark 70. The single-invariant lattice of Theorem 68 can be strictly coarser than $e\mathbb{Z}$ (the smallest example is $(11, 24, 31)$, whose lattice admits $N = 354$), so the single invariant alone does not confine the admissible counts to Zhang's constructions. The mirror invariant of Theorem 69 eliminates every such value (354 by parity, 15689 on $(19, 261, 271)$ already by integrality), and the fully refined spectrum *coincides with Zhang's family on every tile* — verified exhaustively over all 138 primitive triples with $a, b < 400$. The necessary side of the isosceles and F_1 branches is therefore closed. What remains open in these branches is exactly the realizability of the *small* family members. Not all of them are open: a family member that is also a sum of two squares (or 2, 3, 6 times a square) is realized by the commensurable construction [4], so the genuinely undecided members are precisely those with a prime $\equiv 3 \pmod 4$ to an odd power. (Thus $N = 40 = b(a+b)$ on $(3,5,7)$ is realized as 2^2+6^2, and $N = 65 = b(a+2b)$ on $(3,5,7)$ as 1^2+8^2, while $105 = 3 \cdot 5 \cdot 7$ is not a sum of two squares.) Zhang's constructions cover m in certain residue classes from some bound onward, so the sharpest open instances are the smallest such members: $N = 105 = b(a+b) = 7 \cdot 15$ on the F_1 target $(105, 56, 91)$ of the tile $(8, 7, 13)$, then 120, 132, 154, 184. Each has membership undecided by any known necessary condition, and is exactly the kind of bounded existence question the engine of Section 7 addresses — though for N of this size a complete exhaustive search is not yet computationally in reach.

Remark 71. Admissibility is necessary, not sufficient. The smallest admissible value is $N = 33 = 3 \cdot 11$ on the tile $(5, 3, 7)$ (where $d = 3$, $e = 1$, so the divisibility is vacuous); it is nevertheless excluded by Beeson's boundary analysis, which rules out $N < 36$ [3], while Herdt's 2673-tile tiling is the member $w = 9$ of the same family $33w^2$. The parity clause is decisive: on the tile

$(7, 8, 13)$ (where $d = 2$, $e = 2$, $c - a - b = -2$) the divisibility admits all $N = 2w^2 \cdot 23$, but the count $M = -w$ must have the parity of $N = 2w^2 \cdot 23$, which is even — forcing w even, say $w = 2m$; the refined spectrum is exactly $N = 184m^2 = m^2\, b(a + 2b)$ — *precisely* Zhang's constructed family [7] for this tile. In particular $N = 46$, previously the smallest value passing every known necessary condition on this branch, is *excluded* (Section 7 checks the remaining branches), in exact agreement with the prediction of Zhang's conjecture. Zhang's constructions realize $N = m^2 b(a+2b)$ for m in certain residue classes; the refined spectrum bounds the realized set from outside, and Theorem 68 determines it exactly.

Proposition 72 (The other branches)**.** *With the same normalizations: F_1 forces $N = dw^2(a+b)$ where $b = de^2$, d squarefree (from $b \mid k^2$ as above); F_2, F_3, F_4 force $N = N_0 k^2$ exactly, with $N_0 = (a + 2b)(2a + b)$, $3(a + b)(a + 2b)$, $(a + b)(2a + b)$; the tile-similar shape forces $N = n^2$; and an equilateral target of side S forces $ab \mid S^2$, $N = S^2/(ab)$ (so N and ab have the same squarefree kernel) together with $(c + a - b) \mid 3S$ and $(c + b - a) \mid 3S$.*

Proof. The scalene statements are the area identity (6) with the side-vectors of Section 3 (F_1 repeats the $b \mid k^2$ analysis). For the equilateral target the three sides, traversed counterclockwise, differ in direction by $\frac{2\pi}{3} = 2 \cdot \frac{\pi}{3}$, so f takes one sign on all three and $\Phi_f(\partial ABC) = \pm 3S$ for both invariants; Corollary 53 gives the two divisibilities, and the area identity gives $N\, ab = S^2$. □

The equilateral $2\pi/3$ spectrum. The equilateral divisibilities of Proposition 72 reduce to a clean finite criterion.

Proposition 73 (Equilateral admissibility)**.** *Let an equilateral triangle of side S be N-tiled by a $2\pi/3$ tile (a, b, c), and put $X = c + a - b$, $Y = c + b - a$. Then $XY = 3ab$, and $s := 3S/X$, $t := 3S/Y$ are positive integers with*

$$st = 3N, \qquad s \equiv t \equiv N \pmod 2, \qquad (t - s)^2 + 16N = q^2 \;\; \textit{for an integer } q;$$

conversely (s, t, q) determine the instance: (a, b, c) is proportional to $\big(q+(t-s),\, q-(t-s),\, 2(s+t)\big)$, and primitivity pins the scale, hence S.

Proof. $XY = c^2 - (a - b)^2 = 3ab$ by (2). By Proposition 72, $X \mid 3S$ and $Y \mid 3S$, so $s, t \in \mathbb{Z}_{\geq 1}$; indeed $s = M_\alpha$ and $t = M_\beta$ are the two invariant counts, so $s \equiv t \equiv N \pmod 2$ by Corollary 53, and $st = 9S^2/(XY) = 9\, Nab/(3ab) = 3N$. From $X - Y = 2(a - b)$ we get $a - b = \frac{3S}{2}\big(\frac{1}{s} - \frac{1}{t}\big) = \frac{S(t-s)}{2N}$, and $4ab = 4S^2/N$ gives $(a+b)^2 = (a-b)^2+4ab = \frac{S^2}{4N^2}\big[(t-s)^2+16N\big]$; as $a+b$ is an integer, $q := 2N(a + b)/S$ is a rational with integer square, hence an integer, and $(t - s)^2 + 16N = q^2$. Then $a - b$, $a + b$ and $c = (X + Y)/2 = \frac{S(s+t)}{2N}$ give the displayed proportions. □

The equilateral criteria as divisor conditions. Both equilateral square criteria — ours and Beeson's — complete the square.

Proposition 74 (Conic form)**.** *For an equilateral target: a $2\pi/3$ tile requires a factorization $uv = 16N^2$ with $u+v \equiv 0 \pmod 2$ and $\frac{u+v}{2} - 5N = s^2$ for a divisor s of $3N$ with $s \equiv 3N/s \equiv N \pmod 2$; a $\pi/3$ tile requires a factorization $uv = 16N^2$ with $\frac{u+v}{2} = 5N - M^2$ for some $0 < M < \sqrt{N}$, $M \equiv N \pmod 2$. These are equivalent to the criteria of Proposition 73 and of* [5, Thm. 3] *respectively.*

Proof. For the $2\pi/3$ case, with $t = 3N/s$ the condition $(t - s)^2 + 16N = q^2$ becomes, after multiplying by s^2, $(qs)^2 = (3N-s^2)^2+16Ns^2 = (s^2+N)(s^2+9N)$, i.e. $(s^2+5N)^2-(qs)^2 = 16N^2$; factor the difference of squares. For the $\pi/3$ case, $(9N - M^2)(N - M^2) = y^2$ is $(5N - M^2)^2 - 16N^2 = y^2$, i.e. $(5N - M^2 - y)(5N - M^2 + y) = 16N^2$. □

The necessary side of the equilateral branch is thus elementary — a divisor condition on $16N^2$ — and, per N, pins finitely many fully determined instances. Both square-completions of Proposition 74 are formalized and machine-checked in Lean (`lean/EquilateralConic.lean`): the elimination of t giving $(qs)^2 = (s^2 + N)(s^2 + 9N)$, the resulting factorization $(s^2 + 5N - qs)(s^2 + 5N + qs) = 16N^2$, and the $\pi/3$ identity $(5N - M^2)^2 - 16N^2 = (9N - M^2)(N - M^2)$, all over $\mathbb{Z}$. The elliptic curves of Laczkovich [11] concern the *realizability* of such instances, not their enumeration.

The values 14 and 15. The machinery above, combined with the published branch theorems, decides every branch of the classification for $N = 14$ and $N = 15$ by finite computations (`verify_frontier.py`): the commensurable, tile-similar, right-angle and $\gamma = 2\alpha$ branches exclude both values by their known N-forms; the sporadic $2\pi/3$ shapes exclude both by the admissible spectra of Theorems 66 and Proposition 72; the $\pi/3$-tile equilateral case excludes both by Beeson's square criterion ($(9N - M^2)(N - M^2)$ is a perfect square for some $M < \sqrt{N}$ [5, Thm. 3], which fails for every admissible M); and of the five target shapes of the $3\alpha + 2\beta = \pi$ branch [2], four exclude both values through their published tiling equations (checked by exact rational-root enumeration), while the isosceles target with base angles $\alpha+\beta$ admits, for $N = 14$, the single candidate tile $(45, 56, 81)$ with $M = 2$ — which is destroyed by Beeson's boundary congruence $M \equiv -m \pmod{\gcd(a, c)}$ (it forces at least seven b-edges of length 56 on a base of length 140).

What survives everything is four fully explicit finite instances:

N	branch	tile	target
14	equilateral, $2\pi/3$	$(7, 8, 13)$	equilateral, $S = 28$
15	equilateral, $2\pi/3$	$(3, 5, 7)$	equilateral, $S = 15$
15	$3\alpha + 2\beta = \pi$, iso-$(\alpha{+}\beta)$, $M = 1$	$(56, 15, 64)$	$(120, 120, 105)$
15	$3\alpha + 2\beta = \pi$, iso-$(\alpha{+}\beta)$, $M = 3$	$(4, 15, 16)$	$(60, 60, 15)$

In each, the tile is *uniquely determined* (Proposition 73 for the equilateral rows; the tiling equation $N/M^2 = (1 + s)/(1 - s)$ of [2, Thm. 17] for the isosceles rows), so each is a single bounded existence question: can 14 (resp. 15) congruent copies of the stated tile fill the stated triangle?

Each instance was settled by an exhaustive, exact search (`code/engine/`). The engine is an advancing-front algorithm over $\mathbb{Q}(\sqrt{D})$ ($D = 3, 23, 7$; every geometric decision exact, no floating point) whose branching rule is provably complete: at the lexicographically least vertex v of the unfilled region — necessarily convex — every tile of any tiling whose closure contains v has a corner there (an edge through v would subtend angle π), so the clockwise-most of them has a corner at v with one side along the clockwise boundary ray; this leaves at most six placements (three corner types, two adjacent sides each, the two choices being mirror images). An exhausted search is therefore a proof of non-existence. Two prunes are used, each a theorem about completable states: the area of every region component is a positive integer multiple of the tile area; and a maximal straight boundary run with *convex* endpoints has length in $\mathbb{N}a+\mathbb{N}b+\mathbb{N}c$ (whole tile edges cannot pass a convex corner — runs with a reflex endpoint are not pruned, as an edge may continue past a reflex corner into the region). The engine passes positive controls — it finds the $N = 4$ reptile subdivisions in both tile families and a hand-built non-edge-to-edge two-tile configuration with T-junctions, each found tiling re-verified by an independent checker — and, as a robustness check, all four verdicts below are unchanged when the run prune is disabled entirely (node counts without it in parentheses).

instance	tile	target	verdict	nodes
$N = 14$, equilateral	$(7, 8, 13)$	$S = 28$	exhausted, no tiling	133 (482)
$N = 15$, equilateral	$(3, 5, 7)$	$S = 15$	exhausted, no tiling	156 (604)
$N = 15$, iso-$(\alpha{+}\beta)$	$(56, 15, 64)$	$(120, 120, 105)$	exhausted, no tiling	37 (526)
$N = 15$, iso-$(\alpha{+}\beta)$	$(4, 15, 16)$	$(60, 60, 15)$	exhausted, no tiling	2 (10)

Theorem 75. *No triangle can be cut into* 14 *congruent triangles, and none into* 15.

Proof. By the branch sweep above, any 14- or 15-tiling would live in one of the four listed instances, each with its tile and target uniquely determined; the exhaustive searches refute all four. The dependencies are those of Theorem 1 together with [2, Thms. 3, 7, 11, 14, 17, 19], [5, Thm. 3], and the declared computer assistance (exact arithmetic, complete branching, independently validated). ☐

The values 21, 22, 30, and the instance 46. The same sweep, sharpened by the parity clause of Corollary 53, extends further (`verify_frontier.py`). Three remarks carry the new weight. First, the invariant applies to the *scalene* F_1 target: its sides $k(a{+}b), ka, kc$ have $\frac{\pi}{3}$-coefficients $0, 2, 3$ (the same turning computation as Lemma 52), so

$$M_\alpha = \frac{k(2a + b - c)}{c + a - b}, \qquad M_\beta = \frac{k(2a + b + c)}{c + b - a}$$

must be integers of the parity of N. For $N = 21$ this destroys the unique structural candidate, the genuine tile $(5, 16, 19)$ with $k = 4$: $M_\alpha = 28/8 \notin \mathbb{Z}$ — while Zhang's constructed F_1 family $N = m^2 b(a + b)$ passes, so the invariant separates the candidate from the constructions. (For $N = 30$ the candidate $(7, 8, 13)$, $k = 4$ has $M_\alpha = 3$, killed by parity.) Second, for $N = 22$ and $N = 30$ *every* branch dies arithmetically — the two isosceles-$(\alpha{+}\beta)$ candidates of 22 fall to the boundary congruence, those of 30 likewise — so no search is needed at all. Third, for $N = 21$ exactly two instances survive, both isosceles-$(\alpha{+}\beta)$: $M = 1$, tile $(110, 21, 121)$ on $(231, 231, 210)$, and $M = 3$, tile $(10, 21, 25)$ on $(105, 105, 42)$; both live in $\mathbb{Q}(\sqrt{6})$, and the engine exhausts both:

instance	tile	target	verdict	nodes
$N{=}21$, iso-$(\alpha{+}\beta)$, $M{=}1$	$(110, 21, 121)$	$(231, 231, 210)$	exhausted	107 (4266)
$N{=}21$, iso-$(\alpha{+}\beta)$, $M{=}3$	$(10, 21, 25)$	$(105, 105, 42)$	exhausted	15 (102)

Finally, the parity clause settles the value that Remark 71 identified as the sharpest test of Zhang's conjecture: $N = 46$ passes the divisibility spectrum on $(7, 8, 13)$ but not the parity, and the sweep excludes every other branch, so 46 is not realizable — exactly as Zhang's conjecture predicts.

The sweep continues: $N = 38$ and $N = 42$ die in every branch by arithmetic alone (both squarefree, so the $\gamma = 2\alpha$ branch is closed by [3, Thm. 11.7], and no sporadic, equilateral or $3\alpha + 2\beta$ candidate survives), while 33, 35 and 39 each reduce to two isosceles-$(\alpha{+}\beta)$ instances (the sporadic-$2\pi/3$ candidate $(5, 3, 7)$ at $N = 33$ is excluded by Beeson's $N < 36$ boundary analysis [3]); all six searches exhaust:

instance	tile	target	nodes
$N{=}33$, $M{=}1$	$(272, 33, 289)$	$(561, 561, 528)$	489
$N{=}33$, $M{=}3$	$(28, 33, 49)$	$(231, 231, 132)$	49
$N{=}35$, $M{=}1$	$(306, 35, 324)$	$(630, 630, 595)$	523
$N{=}35$, $M{=}5$	$(6, 35, 36)$	$(210, 210, 35)$	2
$N{=}39$, $M{=}1$	$(380, 39, 400)$	$(780, 780, 741)$	853
$N{=}39$, $M{=}3$	$(40, 39, 64)$	$(312, 312, 195)$	67

Theorem 76. *No triangle can be cut into* 21, 22, 30, 33, 35, 38, 39, 42, *or* 46 *congruent triangles.*

Proof. The sweeps above; for 21, 33, 35 and 39, additionally the exhaustive searches listed. Dependencies as in Theorem 75. □

$N = 44$ **is realizable.** The value 44, the smallest undetermined after Theorem 76, has three live branches. Two die: the isosceles-($\alpha+\beta$) instance (tile $(30, 11, 36)$, target $(132, 132, 110)$, from Theorem 17 of [2] at $M = 2$) is exhausted by the engine (15654 nodes, independently in both engine implementations), and the $\gamma = 2\alpha$ branch is excluded by Beeson's Theorem 11.18 [3] ($N \geq 45$ for that branch; his Table 3 lists and eliminates the unique $N = 44$ boundary candidate $(25, 11, 30)$). The third branch is the isosceles-base-β case of [2, Thm. 14]: at $(N, M) = (44, 6)$ the tiling equation gives $s = a/c = \frac{1}{2}$, hence the tile $(2, 3, 4)$ — angles $(\alpha, \beta, 2\alpha+\beta)$ with $3\alpha + 2\beta = \pi$, $\cos\alpha = \frac{7}{8}$, $\cos\beta = \frac{11}{16}$, $\cos\gamma = -\frac{1}{4}$ — and the target $(16, 16, 22)$ with base angles β. Here the exhaustive search does not exhaust: *it finds a tiling*, at node 858163.

Theorem 77. *A triangle can be cut into* 44 *congruent triangles: the isosceles triangle with sides* $(16, 16, 22)$ *is tiled by* 44 *congruent copies of the* $(2, 3, 4)$ *triangle (Fig. 1). Consequently* $44m^2$ *is a tile count for every* $m \geq 1$.

Proof. The tiling is exhibited explicitly (`code/engine/tiling_N44B.txt`: exact coordinates in $\mathbb{Q}(\sqrt{15})$), and its certificate is *machine-verified in Lean 4 with no axioms whatsoever* (`lean/Tiling44.lean`: the kernel checks, by pure computation over $\mathbb{Z}[\sqrt{15}]$ after scaling by 8, that all 44 triangles have squared side multiset exactly $\{256, 576, 1024\} = (8 \cdot (2, 3, 4))^2$, are positively oriented, lie in the closed target via half-plane checks, admit an explicit separating edge-line for each of the 946 pairs, and have signed areas summing exactly to the target's; `#print axioms` reports the theorem depends on no axioms). The bridge from the certificate to the tiling statement is classical: closed half-plane separation gives pairwise disjoint interiors (convexity), the vertices-in-target checks give containment (convexity of the target), and then exact equality of total area forces the union of the closed tiles to be the whole target, since the complement is open with measure zero. So this is an N-tiling in the strict sense — *unconditionally*: this theorem depends on none of the cited classification inputs. The consequence follows by subdividing each tile into m^2 similar copies. □

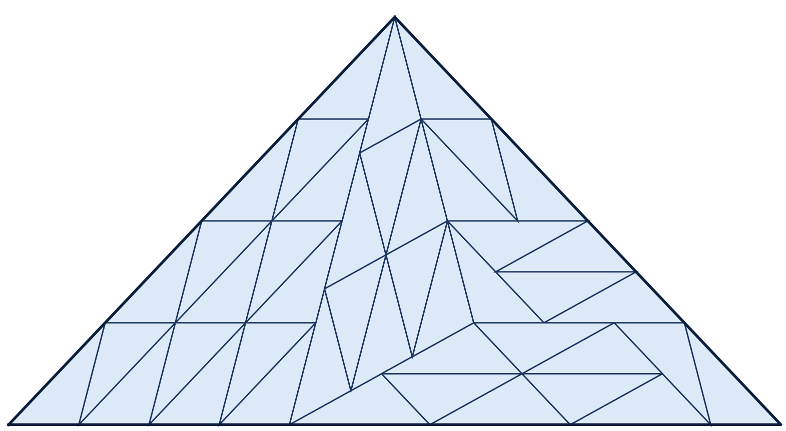

FIGURE 1. The 44-tiling of $(16, 16, 22)$ by the tile $(2, 3, 4)$, found by exhaustive search and verified exactly. To our knowledge the first tiling in any of the three isosceles $3\alpha + 2\beta = \pi$ cases, and the smallest known triangle tiling in any incommensurable branch (the previously smallest was Herdt's 1215).

Three remarks. First, 44 is not a square, not a sum of two squares ($11 \equiv 3 \bmod 4$), and not $2, 3, 6$ times a square: it is the first tile count *outside the commensurable families* to be certified realizable at small size, and the first new member of the tile-count set discovered by these methods — every other frontier value was excluded. Second, Beeson's constructions in [2] cover the two scalene $3\alpha + 2\beta = \pi$ shapes (his triquadratic tilings, smallest $N = 28$ — which, we note, settles 28's membership and is used in our accounting below); the isosceles cases had, to our knowledge, no known tiling, and the smallest known tiling in any incommensurable branch was Herdt's 1215. Third, the value demonstrates that the small members of the $3\alpha + 2\beta = \pi$ tables are not uniformly impossible: each must be searched, and Theorem 82's per-instance decision is genuinely two-sided.

The values 51, 55, 57, 62, and the $\gamma = 2\alpha$ boundary algorithm. The sweep continues past the forties. The values $51 = 3 \cdot 17$, $55 = 5 \cdot 11$ and $57 = 3 \cdot 19$ are squarefree, so the $\gamma = 2\alpha$ branch is closed by [3, Thm. 11.7], and every branch except isosceles-$(\alpha+\beta)$ dies arithmetically; each surviving instance is uniquely determined and the engine exhausts all six (62 dies in every branch outright, as 42 did):

instance	tile	target	nodes
$N{=}51,\ M{=}1$	$(650, 51, 676)$	$(1326, 1326, 1275)$	3649
$N{=}51,\ M{=}3$	$(70, 51, 100)$	$(510, 510, 357)$	180
$N{=}55,\ M{=}1$	$(756, 55, 784)$	$(1540, 1540, 1485)$	5911
$N{=}55,\ M{=}5$	$(24, 55, 64)$	$(440, 440, 165)$	40
$N{=}57,\ M{=}1$	$(812, 57, 841)$	$(1653, 1653, 1596)$	8961
$N{=}57,\ M{=}3$	$(88, 57, 121)$	$(627, 627, 456)$	193
$N{=}63,\ M{=}1$	$(992, 63, 1024)$	$(2016, 2016, 1953)$	15493
$N{=}63,\ M{=}3$	$(12, 7, 16)$	$(84, 84, 63)$	44859
$N{=}63,\ M{=}7$	$(8, 63, 64)$	$(504, 504, 63)$	2
$N{=}69,\ M{=}1$	$(1190, 69, 1225)$	$(2415, 2415, 2346)$	37995
$N{=}69,\ M{=}3$	$(130, 69, 169)$	$(897, 897, 690)$	523

The three $N{=}63$ rows close its isosceles-$(\alpha+\beta)$ branch. The value 63 itself then hangs on its $\gamma = 2\alpha$ instance — the boundary algorithm (below) leaves exactly one candidate, tile $(9, 7, 12)$ with target $(63, 63, 84)$ — and the engine exhausts it too, at 884921 nodes; to our knowledge the first value decided *inside* Beeson's "swath of ignorance" $[45, 1125]$ for the $\gamma = 2\alpha$ branch [3, §11.6]. The values 70 and 78 die in every branch outright, as 42 and 62 did. The value 66 reduces to its $\pi/3$-equilateral instance (tile $(11, 96, 91)$, side 264, coloring number $M = 4$), which Beeson rejects [5, Thm. 7]: the tiling must witness a relation $91j = 11\ell + 96m$ with $91j < 264$, and neither $j = 1$ nor $j = 2$ admits one. And 77 goes the *other* way: its $(2\alpha, \alpha, 2\beta)$ instance satisfies Beeson's second tiling equation at $(M, s) = (5, \frac{1}{2})$ — $25 \cdot \frac{(2-\frac{1}{4})(3-\frac{1}{4})}{(1-\frac{1}{2})^2(2+\frac{1}{2})^2} = 77$ — and his four-component construction [2] realizes every solution, so *a triangle can be cut into* 77 *congruent triangles*: the $(2\alpha, \alpha, 2\beta)$ triangle over the same tile $(2, 3, 4)$ that Theorem 77 uses for 44.

For values that are *not* squarefree the $\gamma = 2\alpha$ branch needs more than Theorem 11.7. Beeson's isosceles paper provides exactly the needed machinery: the tile must be $(k^2,\ m^2{-}k^2,\ mk)$ with $\gcd(k, m) = 1$, $2k > m > k$ [3, Lem. 11.2]; m is bounded in terms of N [3, Lem. 11.12]; the side X and base Y satisfy $X^2 = Nab$, $Y = (m/k)X$ and must decompose as $X = pa + qb + rc$ with $q, r \ge 1$ and $p + q + r < (N{+}1)/4$ [3, Lems. 11.11, 11.13] and $Y = ua + vb + wc$ with $v, w \ge 1$ (the base version of Lemma 11.13, whose pigeonhole proof is stated for the base); and the pair is rejected when every X-representation has $r = 1$ while every Y-representation has $w \in \{1, 2\}$ [3, Lem. 11.17]. We implemented this boundary algorithm independently (`code/gamma2alpha.py`) and calibrated it against Beeson's reported results: it excludes 20, 28 and 44 and leaves 45 surviving via the tile $(4, 5, 6)$, exactly as in [3, Thm. 11.18]; at 36 our filter is strictly weaker than Beeson's direct argument (it reports a surviving boundary representation), so we use it only in the sound direction. Running it on the undetermined values: *the* $\gamma = 2\alpha$ *branches at* $N = 56$ *and* $N = 60$ *are empty* — no possible boundary tiling exists.

Consequently $N = 56$ reduces far: $\gamma = 2\alpha$ is closed by the algorithm, both isosceles-$(\alpha+\beta)$ instances are exhausted by the engine (M=2: tile $(195, 56, 225)$, target $(840, 840, 728)$, 853 nodes; M=4: tile $(45, 56, 81)$, target $(504, 504, 280)$, 89 nodes), and every other branch dies arithmetically — *except* the $2\pi/3$-equilateral instance: tile $(8, 7, 13)$, equilateral target of side 56, from the factorization $(s, t) = (12, 14)$ of Proposition 73. The membership of 56 is thus equivalent to that single equilateral instance — and the engine *exhausts it*: no tiling exists, at 156646 nodes, identically in both independent engine implementations (node-for-node agreement). Likewise $N = 60$ reduces to its equilateral instance alone (tile $(5, 3, 7)$, side 30): its two isosceles-$(\alpha+\beta)$ instances are exhausted as well (M=2: tile $(56, 15, 64)$, target $(240, 240, 210)$, 123003 nodes; M=6: tile $(4, 15, 16)$, target $(120, 120, 30)$, 39 nodes) — and the equilateral instance itself is exhausted at 853731 nodes: $N = 60$ is excluded. (This also settles the m=2 member $X = 30$ of Zhang's equiconstructibility family for $(5, 3, 7)$ by exact arithmetic; Beeson had reported $X < 105$ unrealizable for this tile in unpublished floating-point work.)

Theorem 78. *No triangle can be cut into* 51*,* 55*,* 56*,* 57*,* 60*,* 62*,* 63*,* 69*, or* 78 *congruent triangles.*

Proof. The branch sweeps and the exhaustive searches above; dependencies as in Theorem 75, together with [3, Thm. 11.7] for the $\gamma = 2\alpha$ branch at the squarefree values 51, 55, 57, 69; the boundary algorithm (Lemmas 11.2–11.17 of [3], `code/gamma2alpha.py`) for the $\gamma = 2\alpha$ branch at 56 and 60; the boundary algorithm's single surviving candidate plus its 884921-node exhaustion at 63; and the 7 232 464-node exhaustion at 66 (Remark 9). □

Remark 79 (The value 70 and the unsound $g \mid M$ exclusions)**.** The value $N = 70$ is undetermined. Its single surviving instance is isosceles-α: tile $(6, 5, 9)$, $M = 8$, target $(45, 45, 70)$, which passes the tiling equation *and* side-integrality ($X = 45$, and indeed $(2f + e) = 8 \mid M$); the exclusion rested solely on $g \mid M$ from [2, Thm. 19] ($g = 3 \nmid 8$), whose proof is unsound (Remark 7); Beeson's own Table 5 marks this very instance "?". We therefore regard 70 as *undetermined* pending the direct exhaustive search of this instance. The value 21, whose analogous isosceles-α candidate (tile $(2, 3, 4)$, $M = 5$, target $(12, 12, 21)$) was likewise previously killed only by the unsound $g \mid M$, *is* excluded: the engine exhausts the instance (13 659 nodes, no tiling), so Theorem 76 stands with its dependence on [2, Thm. 19] removed. The same applies to $N = 39$, whose base-β candidate (tile $(12, 7, 16)$ on $(64, 64, 117)$) the engine exhausts in 825 724 nodes: Theorem 76 therefore stands for 39 as well, independently of [2, Thm. 14]. Of the values whose exclusion rested on an unsound divisibility, only 70 remains pending (Remark 79); 59 and 66 have since been exhausted by exact search, at 1 838 175 and 7 232 464 nodes respectively.

Remark 80. The $2\pi/3$-equilateral exhaustions at $N = 56$ (tile $(8, 7, 13)$, side 56) and $N = 60$ (tile $(5, 3, 7)$, side 30) lie strictly inside the range $12 < N < 1215$ that Beeson leaves *open* for this case — he proves only $N \geq 12$ [5, §5] — and, cross-checked to the node in two independent engine implementations (156 646 and 853 731 nodes respectively), are to our knowledge the first rigorous exclusions inside it. The value 56 moreover settles the m=1 case of Zhang's equiconstructibility family for $(8, 7, 13)$: the side-56 equilateral is *not* tileable by $(8, 7, 13)$, exactly as Zhang's Conjecture 2 predicts (his construction realizes $X = 56m$ only for $m \geq 9$). It also closes the m=1 route of his Proposition 9(1) to the F_1 instance $N = 105$: any 105-tiling of $(105, 56, 91)$, if one exists, does not factor through an equilateral 56-block.

The values 76 to 80. The same branch sweep (`verify_frontier.py`) extends the determined initial segment past 75. For $N = 76 = 4 \cdot 19$ the only live branches are $\gamma = 2\alpha$ (excluded by the boundary algorithm of Lemmas 11.2–11.17 of [3], `code/gamma2alpha.py`) and the $3\alpha + 2\beta = \pi$ isosceles-($\alpha+\beta$) target with tile $(90, 19, 100)$ on the isosceles $(380, 380, 342)$, which the engine exhausts (955 474 nodes); every other branch is dead by its N-form. Hence 76 is not realizable. The prime $79 \equiv 3 \pmod 4$ is excluded by Theorem 1; and 77, 80 are realizable — 77 by Beeson's four-component construction [2], and $80 = 4^2 + 8^2$ by the biquadratic (commensurable) tiling [4].

Theorem 81. *No triangle can be cut into 76 congruent triangles. Every $N \le 80$ except $N = 70$ (Remark 79) is thereby determined: the realizable values are exactly 1–13's known members together with $44, 77, 80$ and the standard commensurable and prime $\equiv 1 \pmod 4$ values, and all others are excluded; in particular every $N \le 69$ is determined without exception.*

Proof. The branch sweep and $\gamma = 2\alpha$ boundary algorithm above settle 76; Theorem 1 excludes the prime 79; Theorems 76–78 give the composite exclusions through 78 other than 70; and the four-component and biquadratic constructions realize 77 and 80. □

Decidability. The pattern of the last two subsections is not an accident. Every branch of the classification now reduces, for each fixed N, to a *finite* list of fully determined instances: the commensurable, similar-tile, right-angle and $\gamma = 2\alpha$ branches are decided by closed N-forms; in the sporadic $2\pi/3$ shapes the divisor structure of N (Theorems 66, 68, Proposition 72) leaves finitely many (tile, scale) pairs; the equilateral cases pin the tile from finitely many factorizations (Proposition 73) or coloring numbers ([5, Thm. 2], which determines the tile from (N, M)); and the five $3\alpha + 2\beta = \pi$ shapes have finitely many (M, s) candidates by their tiling equations [2]. Each surviving instance is a bounded existence question, and the corner-anchored search of `code/engine/` decides it in finitely many steps (the branching is provably complete and the depth is N). Hence:

Theorem 82 (Decidability)**.** *It is decidable, given N, whether some triangle can be cut into N congruent triangles (under the same cited inputs as Theorem 1).*

This extends the decidability that Beeson established for equilateral targets [5, Thm. 4] to the whole problem. What decidability does *not* give is a closed-form description of the set. After Theorem 69 and Proposition 74, the necessary side is closed in *every* branch — the admissible values are Zhang's families in the sporadic shapes and explicit divisor-instances in the equilateral cases — so the entire residual openness of the problem is a single question: *which admissible instances are realized*. Zhang's trapezoid constructions realize the sporadic families for $m \ge 9$ in residue classes, and Laczkovich's elliptic-curve analysis [11] governs equilateral realizability; the open kernel is the small members that are not already realized commensurably — sharpest $N = 105$ on the F_1 target of $(8, 7, 13)$ — and each such instance is individually decidable (Theorem 82).

Remark 83 (no invariant decides realizability)**.** The directional invariants are *conserved* by the tiling process: each tile carries value $\pm(c + a - b)$ for f_α (resp. $\pm(c + b - a)$ for f_β), so placing one tile changes Φ_f of the unfilled region by $\pm V$ and the number of remaining tiles by one — in step, so that Φ_f / V stays integral and keeps the parity of the tile count throughout. Hence the only conditions such an invariant imposes are the admissibility conditions of Theorem 69, and *no* additive boundary invariant can distinguish an admissible target from a partially tiled one; none can obstruct an admissible instance, and none prunes the search below its combinatorial size. For the sharpest open instance $N = 105$ we confirmed this computationally: searching all directional weights of period ≤ 4 in the α-winding (reflections included), the only invariants are

f_α ($M_\alpha = 5$) and f_β ($M_\beta = 21$), both admissible; and the vertex-type balance forced by the angle relations $a\alpha + b\beta + c\gamma \in \{\pi, 2\pi\}$ admits 42,282 nonnegative integer solutions. Deciding realizability of an admissible instance therefore lies beyond every linear or coloring invariant; it is a genuinely combinatorial question, which is why the search of Theorem 82 appears to be essential.

Remark 84 (the tiling group collapses as well)**.** The strongest standard combinatorial obstruction is the Conway–Lagarias *tiling group* $T = \langle \chi_v \rangle / \langle$ collinear additivity; tile boundary words $= 1$ in every orientation and reflection $\rangle$; a tileable region R has $\chi(\partial R) = 1$ in T. Writing edge vectors as complex numbers in $K = \mathbb{Q}(\sqrt{3}, i)$ (rotation by α is multiplication by $\zeta_\alpha = (11 + 4\sqrt{3}\, i)/13$, by $\frac{\pi}{3}$ by $(1 + \sqrt{3}\, i)/2$, reflection by conjugation) and reducing to finite quotients $\mathbb{F}_{p^2}$ with $p \equiv 11 \pmod{12}$ — so $\sqrt{3} \in \mathbb{F}_p$ but $i \notin \mathbb{F}_p$ and the plane stays genuinely two-dimensional — the tile relators, *together with* the commutators [class-1(tile), generator] that normal closure forces (a tile word's class-1 image is itself a relator, hence not central), fill the entire commutator layer. So T collapses to its rank-two abelianization (M_α, M_β): the class-2 quotient is trivial while the class-1 layer keeps rank 2. As $N = 105$ is admissible, $\chi(\partial_{105}) = 1$, and the tiling group does not obstruct it. We verified this exactly for $p \in \{23, 47, 59, 71, 83, 107, 131\}$ (`verify_tiling_group.py`), and — extending to the free nilpotent quotient through class 4 and to a range of finite non-abelian groups (through S_5, $\mathrm{GL}_2(\mathbb{F}_p)$, $\mathrm{PSL}(2,7)$ and the relevant Frobenius groups, which carry genuine non-abelian data: sensitivity controls fire on 92–96% of tile-consistent representations) — found *no* tile-consistent quotient sending ∂_{105} off the identity, with the reptile controls $N = 4, 9$ trivial throughout and a Φ-nonadmissible loop not. The no-go of Remark 83 therefore extends past the linear invariants to the full tiling group: on this tile the tiling-group obstruction is exactly Φ, so deciding realizability of $N = 105$ lies outside every standard tiling invariant.

State of the full problem. Assembling: the commensurable branch is completely determined (squares, sums of two squares, and $2, 3, 6$ times squares, all realized [4]); the similar-tile and right-angle branches are settled in [1, 12]; the primes are settled by Theorem 64. What remains open for the full determination of the tile-count set is: sufficiency in the sporadic $2\pi/3$ branches (Zhang's completeness conjecture [7]; the admissible spectra above are the outer bounds), the $3\alpha + 2\beta = \pi$ and $\gamma = 2\alpha$ branches beyond the partial results of [2, 3], and the equilateral targets for incommensurable tiles, where Laczkovich has connected the possible N to rational points on elliptic curves [11]. With 7 and 11 excluded by [4], the primes by Theorem 64, $14, 15, 21, 22, 30, 33, 35, 38, 39, 42, 46$ by Theorems 75 and 76, $51, 55, 56, 57, 60, 62, 63, 66, 69, 78$ by Theorem 78 (70 pending, Remark 79), 44 *realizable* by Theorem 77, 77 realized by Beeson's four-component construction [2], 28 realized by his triquadratic tilings [2], and the remaining values commensurably realized $(45, 48, 49, 50, 52, 53, 54, 58, 61, 64, 65, 68, 72, 73, 74, 75)$ or prime $(47, 53, 59, 67, 71)$, the membership of *every* $N \leq 75$ is now determined. The value $N = 76$ is likewise excluded ($\gamma = 2\alpha$ empty by the boundary algorithm; its one isosceles-($\alpha+\beta$) instance, tile $(90, 19, 100)$ on $(380, 380, 342)$, exhausted by the engine at 955 474 nodes); with 79 prime (Theorem 1) and $80 = 4^2 + 8^2$ realized, *every* $N \leq 80$ *except the pending* 70 is now determined (Theorem 81).

Standing of the inputs. The external inputs to Theorem 1 and their publication status are as follows: Laczkovich's classification papers [9, 10] and the rep-tiling theorem of Snover–Waiveris–Williams [12] are refereed journal articles, and Beeson's isosceles paper [3] is accepted (to appear in *Integers*); Beeson's remaining branch papers [1, 2, 4, 5] and the Beeson–Zhang rationality theorem [6], which is load-bearing for the integrality of the tile, are arXiv preprints at the time of writing. Zhang's constructions [7] are cited for context only and carry no logical weight here. The parts original to this paper (Sections 3–5 and 7, together with the shape enumeration of Section 2) are self-contained given those inputs, with the arithmetic layer machine-checked (Section 8).

8. Verification

The finite and numerical claims are checked by the scripts in the repository (exact arithmetic). `verify_shapes.py` reproduces the eleven-shape enumeration of Section 2 and the values N_0 of Section 3. `verify_invariant.py` confirms Lemma 52 ($C_f(t) = \pm(c + a - b)$ over all orientations of T), Lemma 51 (both sides agree on explicit subdivision tilings, and (7) cancels at a non-edge-to-edge incidence), and Lemma 54 (no counterexample among primitive 120°-triples up to $c \approx 9 \times 10^6$). `verify_spectrum.py` checks Section 7: the two-positive-squares decomposition for all primes $p \not\equiv 3 \pmod 4$ below 10^5; the scale structure $b \mid k^2 \iff de \mid k$; the admissible spectrum of Theorem 66 (no prime is admissible; 33 and 46 are); and that every instance of $e \mid (a + b - c)$ with $a, b < 3000$ arises from the j-classification of Proposition 67, with none for $d = 1$. As a positive control, both necessary conditions of Theorem 55 hold on the genuine 2673-tile isosceles tiling of Herdt [3]: for its tile $(5, 3, 7)$ and scale $k = 27$, $N = k^2(a + 2b)/b = 2673$ and $M = k(2b + a - 2c)/(c + a - b) = -9$ are integers, as they must be on any true tiling.

The entire *arithmetic and combinatorial* layer of the proof is formalized and machine-checked in Lean 4 with Mathlib (`lean/Erdos634.lean`, twenty-three theorems); every theorem depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`:

> `shape_enumeration`: the eleven-triple completeness of Section 2 — realizable corner types with $\sum m_i = 0$, $\sum k_i = 3$ are exactly the eleven listed — as pure integer arithmetic;
> `k_not_dvd_sum_sub`, `M_not_int`: Lemma 54 ($(c - a - b)/\sqrt{b} \notin \mathbb{Z}$);
> the full arithmetic of Theorem 55 in one master statement: no prime N satisfies the 120°-relation, the area equation $Nb = k^2(a + 2b)$, and the integrality $(c + a - b) \mid k(2b + a - 2c)$ simultaneously (`iso_reduction_identity`, `prime_count_forces_scale`, `no_prime_isosceles_count`);
> `add_not_prime`, `not_prime_of_two_le`, `F1_count_not_prime`–`F4_count_not_prime`: the scalene counts of Proposition 50 are composite ($a + b$ is never prime for a primitive 120°-triple, and the F_2, F_3, F_4 counts factor);
> `prime_three_mod_four_excluded`: a prime $p \equiv 3 \pmod 4$ with $p > 3$ is neither a square, a sum of two squares, nor $2n^2$, $3n^2$ or $6n^2$ (the commensurable branch of Section 2), via Fermat's two-squares theorem;
> `prime_sum_two_pos_squares`: a prime $p \not\equiv 3 \pmod 4$ is a sum of two *positive* squares (the achievability half of Theorem 64);
> `iso_admissible`: Theorem 66 — given $b = de^2$ with d squarefree, the area equation and Φ-divisibility force $k = dew$, $N = dw^2(a + 2b)$ and $e \mid w(c - a - b)$;
> the finite arithmetic of the frontier sweeps, in eight theorems: the $\pi/3$-equilateral square criterion fails at 14 and 15; the first tiling equation has no admissible solution there; the two equilateral factor-pair uniqueness statements; the boundary congruences that kill the isosceles candidates at 14 and 22; the F_1 invariant kill at 21; and the parity kill of 46.

What is *not* formalized is exactly the geometry: the direction grid of Section 2 and Lemmas 51 and 52, which are proved in the text and checked numerically; Mathlib has no theory of planar dissections.

Acknowledgements

This work was carried out with substantial assistance from an artificial-intelligence system (Anthropic's Claude), which proposed the signed-direction invariant, performed the symbolic

and numerical verification, found the elementary proof of Lemma 54, and produced the Lean formalization, under the author's direction and review. The argument depends on the cited classification of Laczkovich and Beeson for the branches other than the $2\pi/3$ tile, and on the rationality theorem of Beeson and Zhang; the shape list coincides with Zhang's six sporadic types [7]. The present contribution is the invariant of Section 4, the consequent exclusion of *every* prime for the isosceles $2\pi/3$ branch (where Beeson had ruled out only $N < 36$ and left primality open), and the self-contained reduction of the scalene types. Its arithmetic layer is machine-checked (Section 8); the two geometric lemmas rest on written proofs supported by numerical checks, and the whole is offered for independent refereeing.

References

[1] M. Beeson, *Triangle tiling I: The tile is similar to ABC or has a right angle*, arXiv:1206.2231.
[2] M. Beeson, *Triangle tiling III: the case* $3\alpha + 2\beta = \pi$, arXiv:1206.2229.
[3] M. Beeson, *Tilings of an isosceles triangle*, arXiv:1206.1974; to appear in Integers.
[4] M. Beeson, *No triangle can be cut into seven congruent triangles*, arXiv:1811.09723.
[5] M. Beeson, *Tiling an equilateral triangle*, arXiv:1812.07014.
[6] M. Beeson and Y. X. Zhang, *Rationality of certain triangle tilings*, arXiv:2604.01314.
[7] Y. X. Zhang, *Tiling triangles with* $2\pi/3$ *angles*, arXiv:2512.22696.
[8] M. Beeson, M. Laczkovich and Y. X. Zhang, *Solution of Erdős Problem 633*, arXiv:2604.03609.
[9] M. Laczkovich, *Tilings of triangles*, Discrete Math. **140** (1995), 79–94.
[10] M. Laczkovich, *Tilings of convex polygons with congruent triangles*, Discrete Comput. Geom. **48** (2012), 330–372.
[11] M. Laczkovich, *Rational points of some elliptic curves related to the tilings of the equilateral triangle*, Discrete Comput. Geom. **64** (2020), 985–994.
[12] S. L. Snover, C. Waiveris and J. K. Williams, *Rep-tiling for triangles*, Discrete Math. **91** (1991), 193–200.
[13] A. Soifer, *How does one cut a triangle?*, 2nd ed., Springer, 2009.